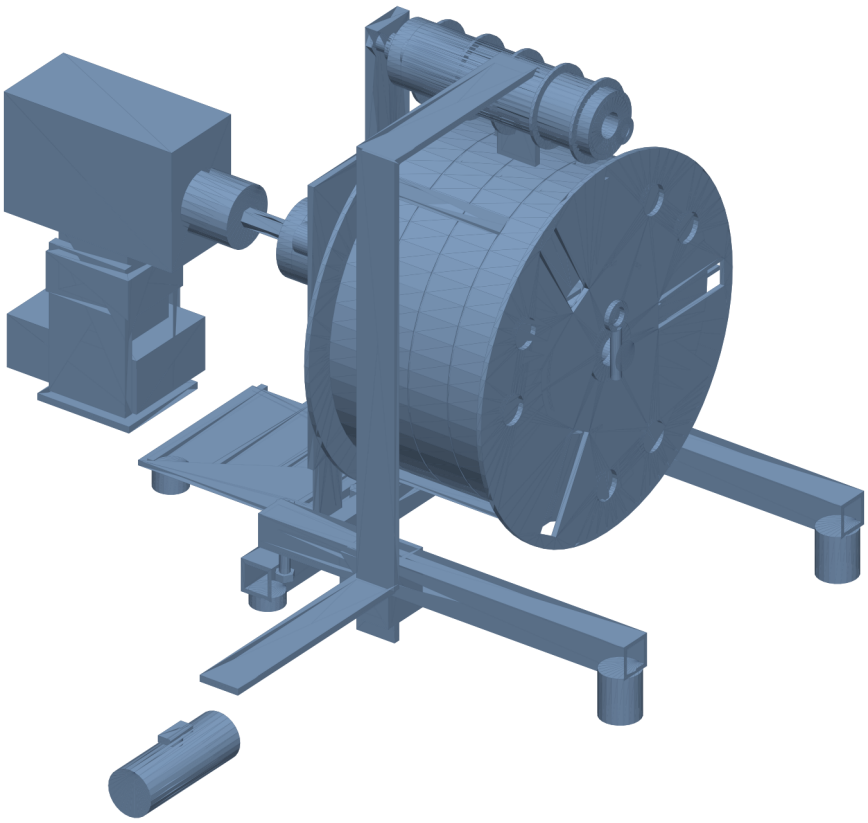
PROJECT PORTFOLIO · DESIGN, CAD & DOCUMENTATION



Navneeth Compact 40M

Portable, drill-driven winder for a single 40 m flat drip tape

Full design documentation · Revision 2.0 · 3D CAD, technical drawings, calculations, bill of materials, assembly and validation status

Navaneeth

Student builder · Linux · IoT · CAD & documentation

Project date 17 Oct 2026 · 50 pages

About Me

Navaneeth · Aspiring Cyber Security Engineer · student builder

Hi, I'm Navaneeth, a Class 12 Computer Science and Maths student and an aspiring **Ethical Hacker** and **SOC Analyst**. I learn by building real systems: a headless Linux home server, a Pi-hole DNS filter, a Telegram bot that controls the server, and Raspberry Pi / ESP8266 home-automation projects.

This document shows another side of the same habit: designing and documenting a physical machine, the **Navneeth Compact 40M**, a drill-driven winder for a 40 m drip tape. I treated it like a system I had to understand end to end, and I recorded every conflict I found in my own first version and how the numbers solved it.

Where I work and what I use

Area	Tools and projects (from my portfolio site)
Systems & networking	Ubuntu Server, systemd, SSH key authentication, UFW firewall, DNS, DHCP, Samba, Nginx
Projects	GatePulse smart gate light (Raspberry Pi, Home Assistant, ESP8266, ESPHome, MQTT) · Home server · Pi-hole (blocks ~31% of DNS queries) · Telegram remote server bot with AI layer · AI CCTV prototype (Docker, RTSP, ONVIF, Frigate)
Programming	HTML/CSS/JavaScript, Python (basic), Bash, Git, Android/Kotlin
Design & CAD (this document)	Parametric CAD scripts (CadQuery, OpenSCAD, Python), STEP/STL/GLB models, drawings, Excel calculation workbook, bill of materials

Education and certificates

12th Std (Science & CS)	Govt. Higher Secondary School, Thuduppathi · expected 2026–27
10th Std (TN State Board)	Govt. High School, Kaikolapalayam · 409 / 500 (81.8%)
Generative AI Mastermind	Outskill · completed May 2026 (LLMs, prompt engineering, API integration)
Web Designing, grade A+	Apollo Computer Education Ltd. · completed Nov 2023

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Languages	English (fluent), Tamil (native)

Profile details taken from my portfolio site, neoxey.github.io/portfolio.

Contents

Everything in this document is taken from the project package: CAD files, drawing sheets, calculation workbook and reports.

About me . . . 2
1 Project overview . . . 4
2 Requirements and solution at a glance . . . 5
3 How it works: tape path and kinematics . . . 6
4 Coordinate system and overall dimensions . . . 7
5 CAD overview . . . 8
6 Rendered views . . . 9
7 Technical drawing sheets (original, 6 sheets) . . . 10
8 Design evolution v1.0 → v2.0 . . . 16
9 Conflict audit . . . 18
10 Geometry audit v1.0 vs v2.0 . . . 19
11 Drum geometry and winding capacity . . . 20
12 Capacity sensitivity . . . 21
13 Drive, speed and winding time . . . 22
14 Torque, tape pull and reaction forces . . . 23
15 Strength verification . . . 24
16 Bearings, shafts and fits . . . 25
17 Mass budget . . . 26
18 Part catalogue . . . 27
19 Bill of materials . . . 36
20 Fastener register . . . 38
21 Assembly instructions . . . 40
22 Strapping and coil removal . . . 42
23 Operation and safety notes . . . 43
24 Maintenance, materials and standards . . . 44
25 Validation matrix . . . 45
26 Open risks and mitigation . . . 46
27 Proposed prototype test plan . . . 47
28 Manufacturing summary . . . 48
29 Digital deliverables and CAD source . . . 49
30 Portfolio highlights and contact . . . 50

1 Project overview

What the machine does and why it is built this way

The Navneeth Compact 40M is a portable field machine for one job: after the irrigation season, wind **one continuous 40 m ALLWIN GOLD flat drip tape** (25 mm lay-flat width, 16 mm inflated) onto a drum, strap the coil in three places, and slide it off in one piece while the valve end stays connected. A cordless drill turns the spindle; the frame carries everything, and a single guide roller feeds the tape across four lanes so the coil stays inside the flanges.

Where it started

The first concept was a single-side “cassette” reel: remove the outer frame, carry everything on one rigid back plate, cantilever the shaft, make the drum and front plate a removable unit and use one fairlead instead of two guide rollers. That concept cut the envelope from about 18.8 L to about 14 L (−25%) and the part groups from 8 to about 5. It also named the limit on compactness: the tape volume itself cannot shrink, so the savings must come from frame, base and protruding parts. The open question at the end of that concept (hose type and diameter) was answered in v2.0 by designing around a flat drip tape.

Design in one paragraph

A 110 mm wide drum with a 2° taper (radius 52.00 mm at the back, 48.15 mm at the front) sits between two Ø220 mm flanges. The taper lets the finished coil slide off along +X, and the 48.15 mm front radius still respects the 48.0 mm minimum bend radius of the tape. The tape is laid in **four 25 mm lanes of 10 m each**, which gives a coil radius of 96.58 mm and leaves 13.42 mm of margin inside the flange. A single lane would need 172.58 mm and overflow the flange by 62.58 mm, so the traversing guide is not optional.

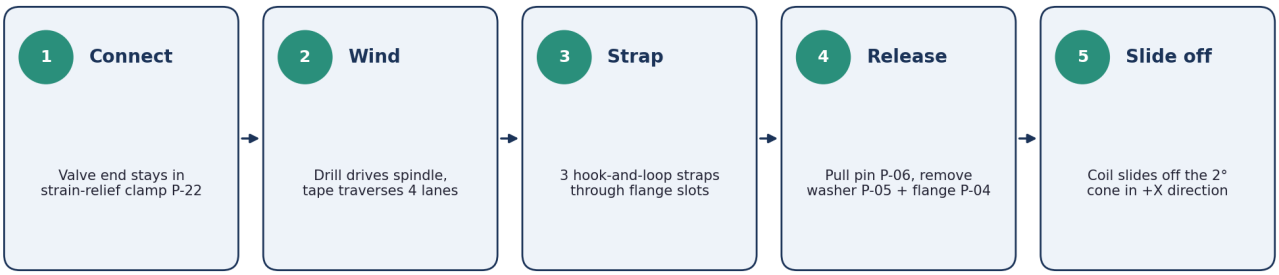


Figure 1 · Operating sequence: connect, wind, strap, release, slide off.

Key numbers

40 m	Ø193.2 mm	86.82	3.097 kg	SF 5.8 – 19.8	+X
tape length, one piece	finished coil	spindle turns per 40 m	dry machine mass (limit 5.0)	verified safety factors	tool-free removal direction

2 Requirements and solution

Fixed requirements from the specification and how revision 2.0 answers each one

Requirement	Target	Solution in v2.0	Result
Tape capacity	1 continuous tape, L = 40 000 mm, width 25.0 mm, t_eff 1.5 mm, R_bend,min 48.0 mm	4-lane traverse (4 × 10 m) on a conical drum, L = 110 mm, R_core 52.00 → 48.15 mm; R_coil = 96.58 mm < R_flange 110 mm	Met · 13.42 mm radial reserve
Stationary valve end	Valve end stays connected and stationary, 0 N pull at the valve	Frame-fixed strain relief and parking clamp P-22 (X 42…74, Y −120) takes 100% of tape force; operating rule documented (RISK-01)	Clamp validated · field test open
Strapping & removal	Strap the coil before removal, pull off in +X without cutting the tape	3 radial slots 16×58 in P-01 and P-04 (0°/120°/240°) + 3 drum channels 16×6 in P-02; quick release via pin P-06, washer P-05, flange P-04	Solved geometrically
Drive & winding time	INGCO CIDLI12206 (12 V, 0–400 / 1500 rpm, 20 Nm); target 3–5 min; torque reaction supported	Form-fit 3/8" stub + M6 cross bolt F-21; frame-fixed torque support P-21 (235.3 N, SF 8.16). A 12:1 … 20:1 gear gives 3–5 min (21.7 rpm nominal)	Support validated · gearbox mandatory (RISK-02)
Mass	≤ 5.00 kg without drill, battery and tape	Material-based CAD mass balance: 3.097 kg dry (4.537 kg with 40 m tape)	Met · 1.903 kg reserve

Project metadata

Revision	2.0 (revised from v1.0)
Date	17 Oct 2026
Units	mm, kg, N, Nm
Drive	INGCO CIDLI12206, 12 V, up to 20 Nm
Bearings	2× SKF 6003-2RS (spindle), 2× SKF 6002-2RS (guide roller)
Material	EN AW-6061-T6 frame and flanges · C45E +QT shafts · POM-C guide parts · NBR 70 feet
Inventory	24 BOM positions P-01 … P-23B · 34 fasteners F-01 … F-34 · 11 joints JNT-01 … JNT-11
Release status	CAD and calculation validated · prototype build released · series release waits on 2 open gates

Status in plain words. The design is fully dimensioned, collision-checked in CAD and analytically verified. It has **not** been physically tested. Two points must be closed before any series release: the kinematic rule for the stationary valve end (RISK-01) and the speed-reducing gearbox between drill and spindle (RISK-02). Both are explained on page 46.

3 How it works

Tape path, drive train and the rule that keeps the valve load-free

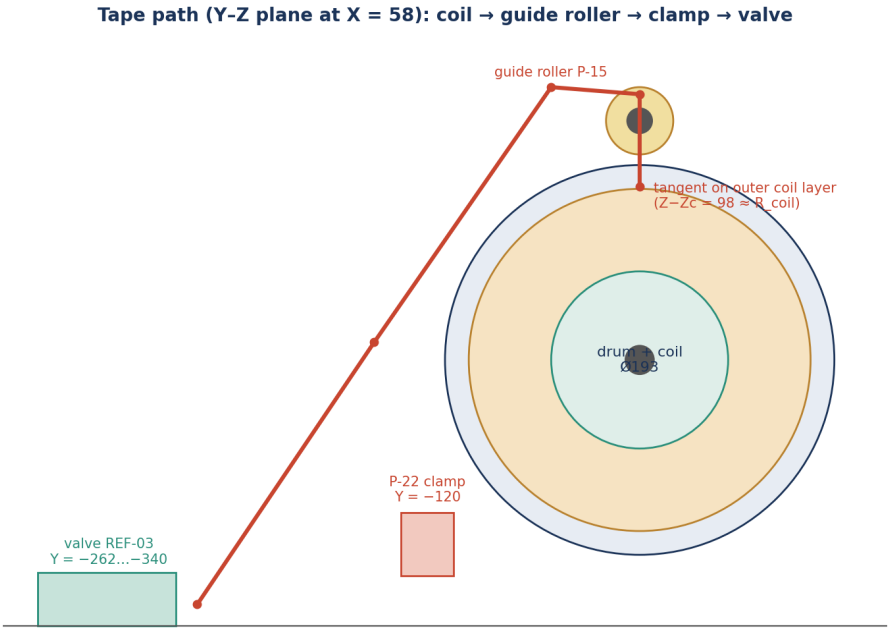


Figure 2 · Tape path in the Y–Z plane at X = 58 mm, drawn from the specification control points [58,0,248] → [58,0,300] → [58,−50,304] → [58,−150,160] → [58,−250,12].

The tape leaves the outer coil layer tangentially (Z − Zc = 98 mm, close to R_coil = 96.6 mm), passes over the guide roller P-15 at Z = 285 mm, runs through the strain-relief clamp P-22 and reaches the stationary valve without tension.

Why the valve end needs a rule

If the tape end were rigidly fixed to the rotating drum core *and* to the stationary valve, the section in between would be twisted 86.82 times, once per spindle revolution. The documented working mode therefore starts at the free field end of the tape, clamped in the drum core slot, and winds toward the valve. The last 0.8 m loop is placed into clamp P-22 after the drum has stopped. The alternative is a rotary union. This is the first of the two open risks.

Drive train

- Drill chuck → 3/8" hex stub on spindle P-07 → form-fit torque transfer through the M6×50 cross bolt F-21 at X = −9 mm into hub P-01 and drum P-02.
- Spindle runs in two SKF 6003-2RS bearings, 50 mm apart, inside stepped housing P-09 on chassis plate P-11.
- The guide roller P-15 turns on two 6002-2RS bearings and follows the tape: ω_roller = ω_spindle × (R_coil / 19.0 mm).
- Reaction torque of the drill (up to 20 Nm, 235.3 N at the handle) goes into the frame through support P-21, not through the user’s wrist.

4 Coordinate system and overall dimensions

All parts, bores, fits and fasteners are referenced to one right-handed system

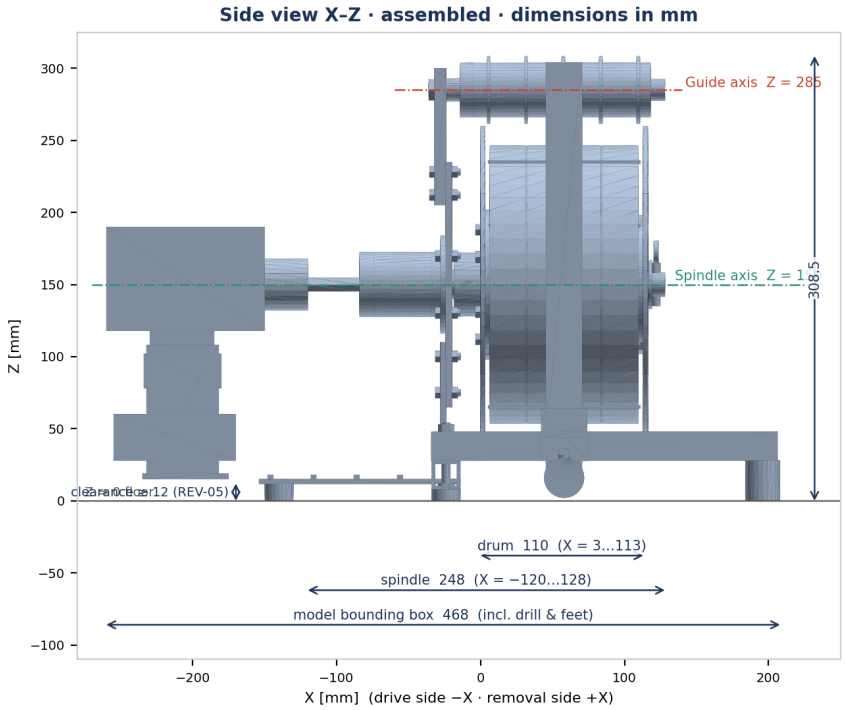


Figure 3 · Side view generated from the STL assembly. The X axis is the spindle axis: drive side at −X, tape removal at +X.

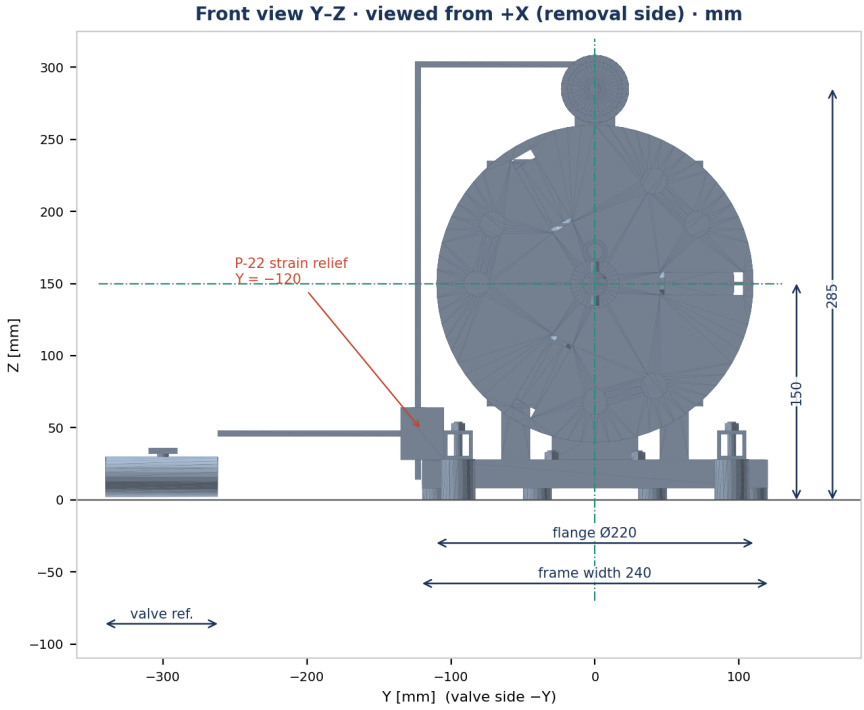


Figure 4 · Front view from +X with the valve side at −Y. Dimensions measured on the assembly model.

Origin X = 0: front face of rear flange P-01 · Y = 0: spindle centre line · Z = 0: footprint plane of the elastomer feet. The spindle sits at exactly Z = 150.0 mm, the guide axle at Z = 285.0 mm.

5 CAD overview

Four standard views of the complete parametric model (original figure from the engineering report)

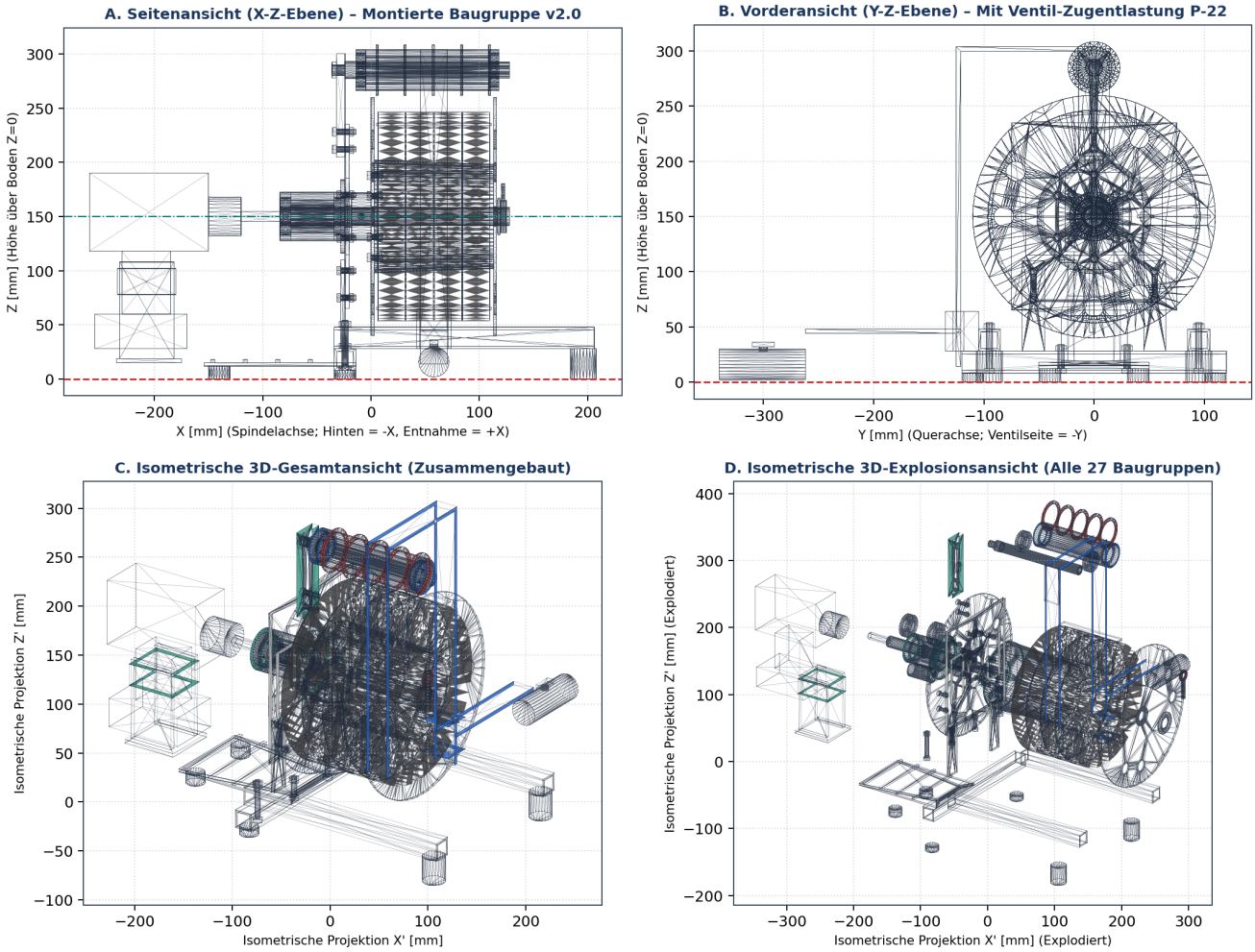


Figure 5 · Panels: A side view X–Z · B front view Y–Z with valve strain relief P-22 · C assembled isometric · D exploded isometric with all 27 positions. Reproduced unaltered from the engineering report.

The model is available as STEP (AP214, assembly plus 27 part files), binary STL (assembly plus 27 parts) and two GLB scenes (assembled and exploded). It can be rebuilt from the parametric scripts listed on page 49.

6 Rendered assembly

Shaded views rendered directly from the STL assembly

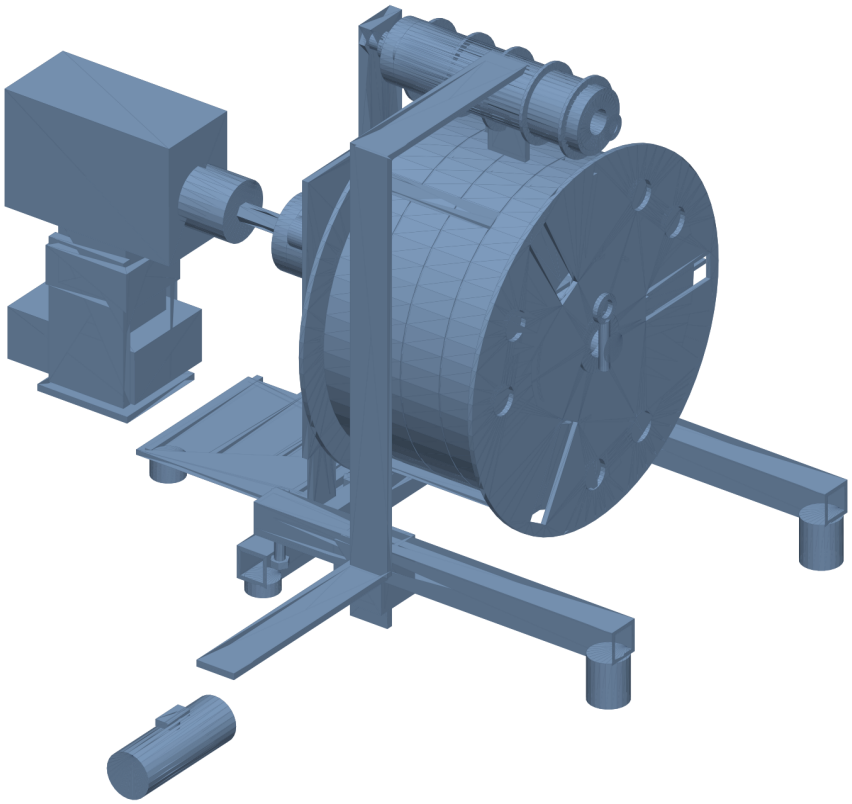


Figure 6 · Isometric view from the front right. The coil sits on the conical drum between the two Ø220 flanges, the guide roller bridges the top, and the drill is held by the torque support on the left.

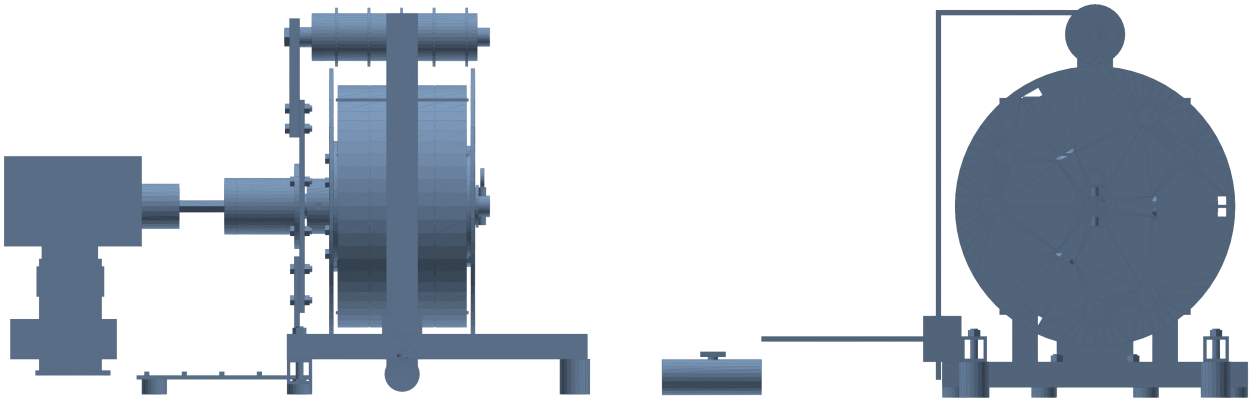
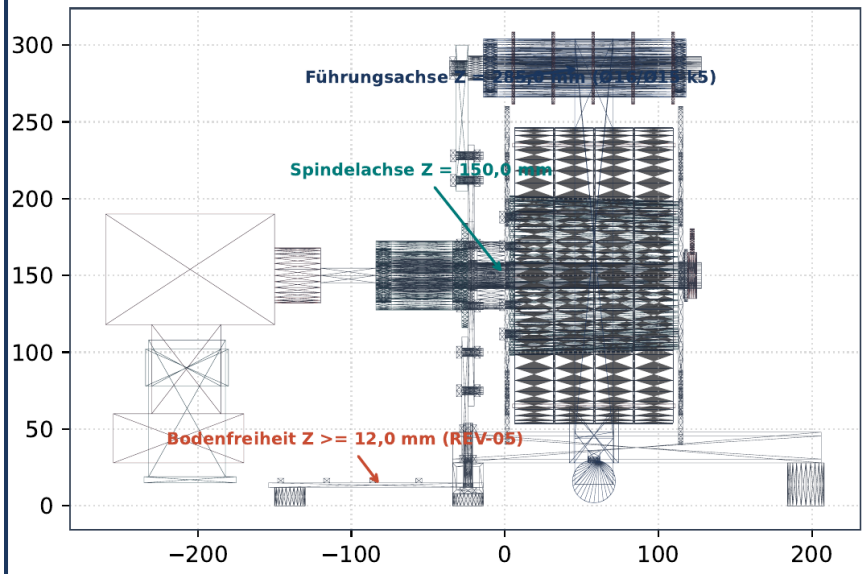
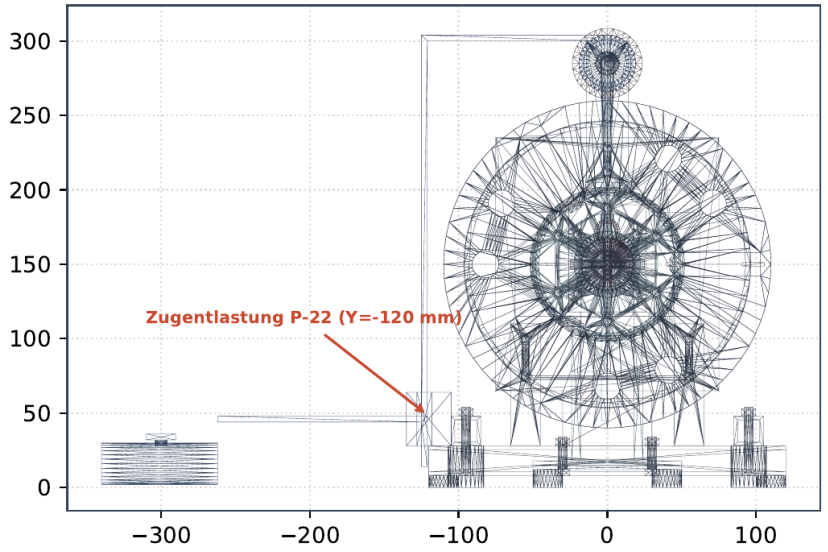


Figure 7 · Left: side view X–Z. Right: front view Y–Z.

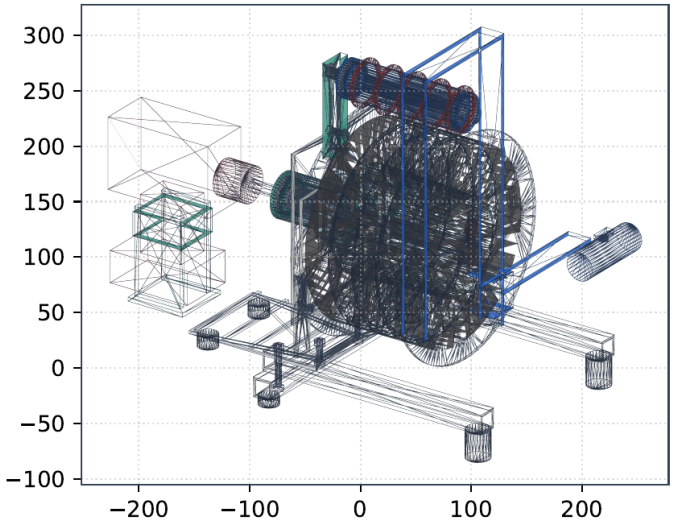
Seitenansicht X-Z (Gesamtlänge 466 mm, Höhe 304 mm)



Vorderansicht Y-Z (Rahmenbreite 240 mm, Flansch Ø220 mm)



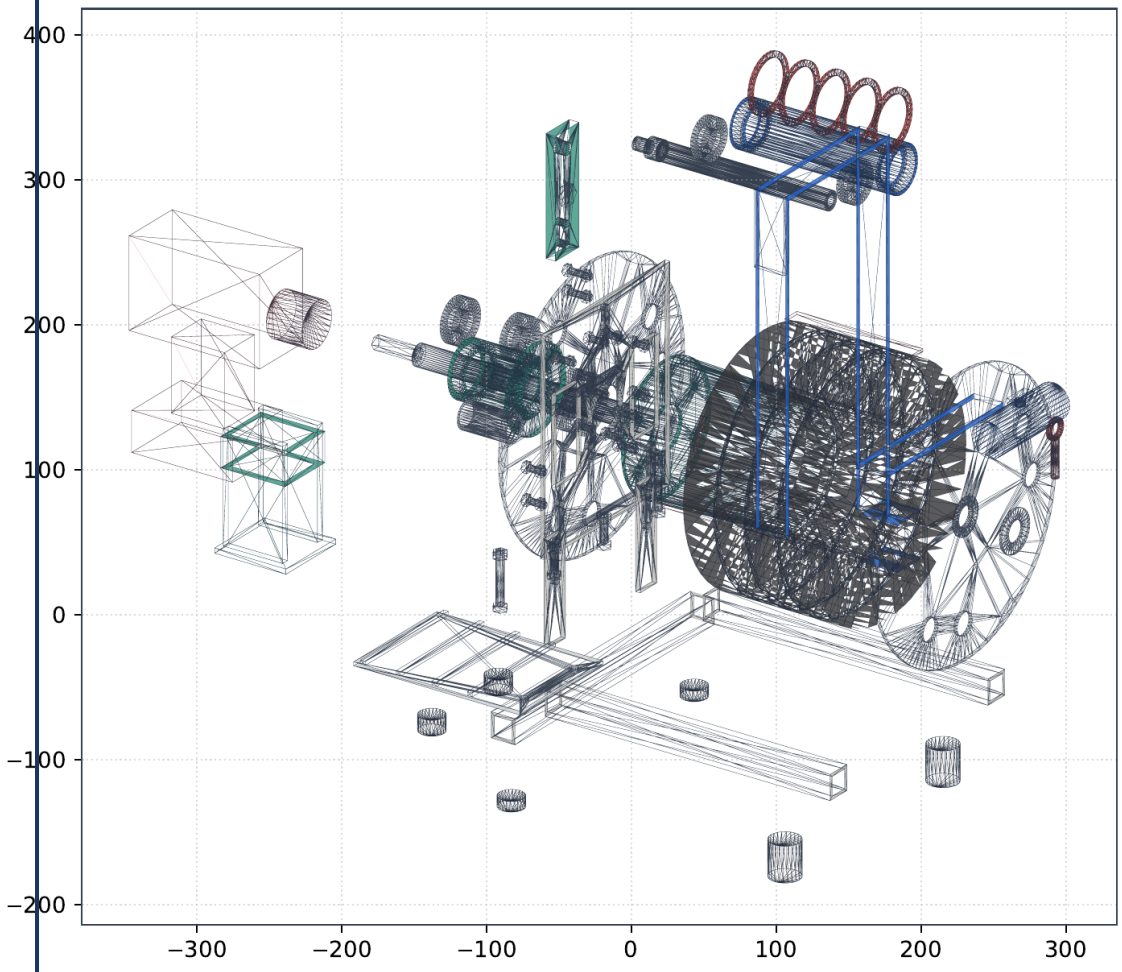
Isometrische Baugruppenansicht (Zusammengebaut)



HAUPTMASSE & BAUGRUPPEN-SPEZIFIKATION (REV 2.0):
- Koordinatenursprung: X=0 (Vorderseite Hinterflansch P-01), Y=0 (Spindelmitte), Z=0 (Bodenebene
- Spindel-Drehachse: Y = 0,0 mm, Z = 150,0 mm | Kragarm-Führungsachse: Y = 0,0 mm, Z = 285,0
- Hauptlagerung: 2x SKF 6003-2RS (17x35x10 mm) bei X=[-84..-74] und X=[-34..-24] in Gehäuse Ø3
- Führungsrolle: 2x SKF 6002-2RS (15x32x9 mm) bei X=[-14..-5] und X=[109..118] auf Achse Ø15,0(
- Drehmomentübertragung: Formschnittige M6x50 Querschraube FK 8.8 bei X=-9,0 mm + Stützbock
- Trockenmasse Maschine (ohne Bohrschrauber, Akku & Tropfband): 3,097 kg (Ziel <= 5,000 kg erfül

PROJEKT: NEox Compact 40M v2.0 | Gesamtbaugruppe NEox Compact 40M v2.0		
Werkstoff: Aluminium 6061-T6 / Stahl C45E / ISO M-E	Masse: 3,097 kg	Maßstab: 1:5
Zeichnungs-Nr.: DRW-ASSY-001-v2	Toleranz: ISO 2768-mK	**Blatt 1/6**

Explosionszeichnung mit Baugruppen-Vektoren und Positionsnummern



VALIDIERTE ENTNAHMESEQUENZ (+X):
1. 3x Klett-Umreifungsbänder (P-03) durch die 3 Flanschschlitze (0°/120°/240°, B=16 mm) und Trommelkanäle (T=6 mm) führen und die 40-m-Rolle (REF-01) fest umreifen.

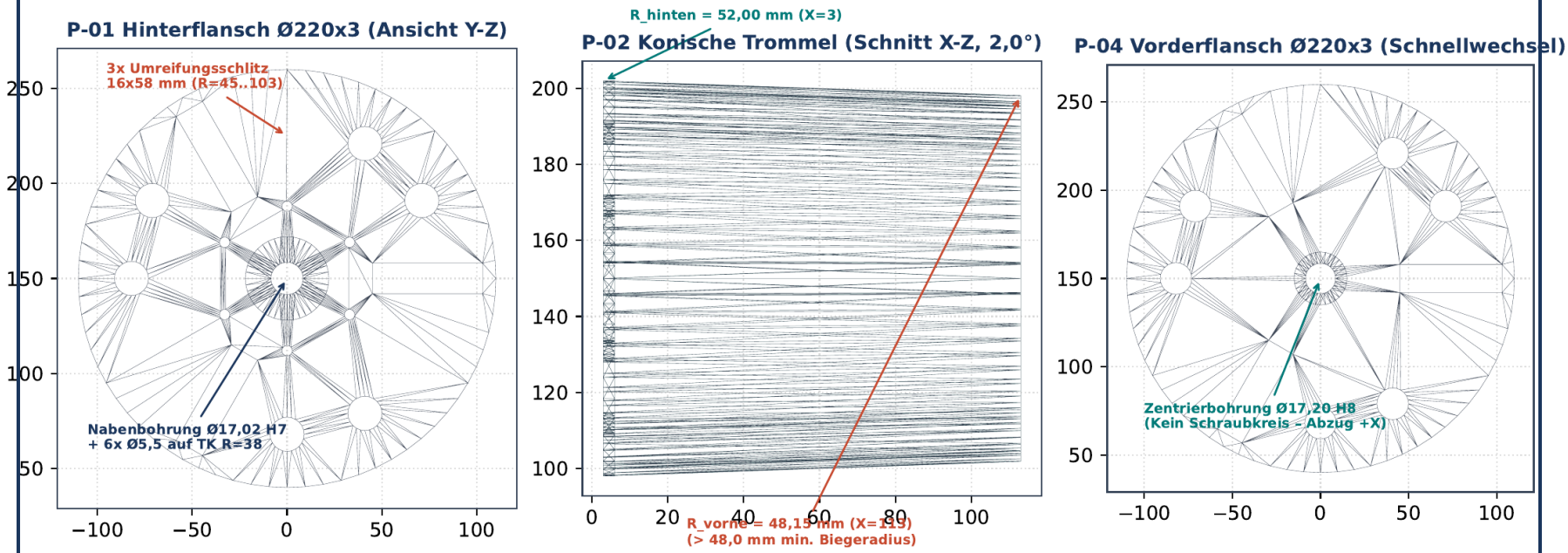
2. Sicherungspin (P-06, Ø6 mm bei X=122 mm) am Zugring in +Z-Richtung herausziehen.

3. Axiale Haltescheibe (P-05, Ø34/Ø17,2x2 mm) in +X-Richtung von der Spindel abziehen.

4. Vorderen Schutzflansch (P-04, Ø220 mm, Zentrierbohrung Ø17,2 H8) in +X abziehen.

5. Umreifte 40-m-Bandrolle (REF-01) über den 2,0°-Trommelkonus (R=52,0 -> 48,15 mm) in +X abziehen, während das Bandende in Zugentlastungsklemme P-22 gesichert bleibt.

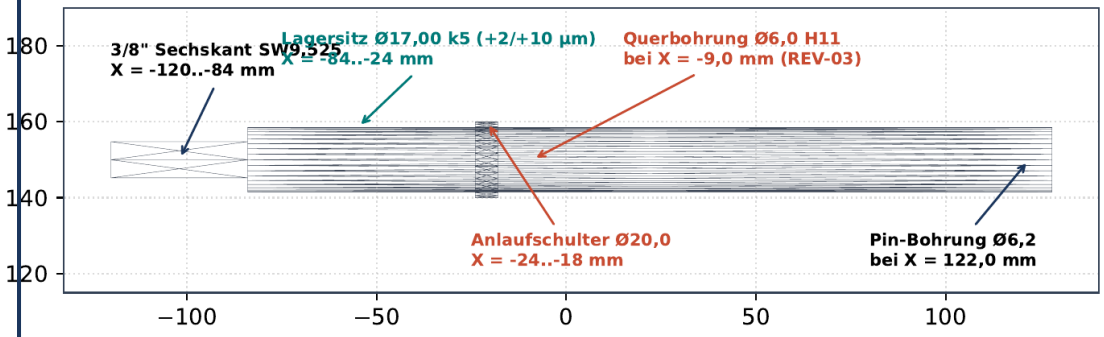
PROJEKT: NEox Compact 40M v2.0 | Explosionszeichnung & Rollen-Entnahmeseque		
Werkstoff: Baugruppe komplett (27 Positionen)	Masse: 3,097 kg	Maßstab: 1:5
Zeichnungs-Nr.: DRW-EXPL-002-v2	Toleranz: ISO 2768-mK	**Blatt 2/6**



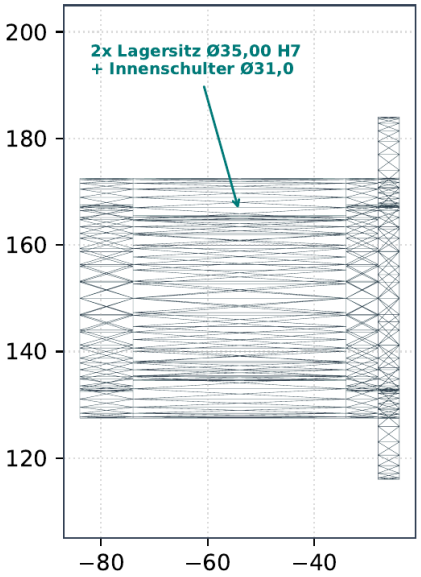
FERTIGUNGSSPEZIFIKATION ROTIERENDE TROMMELBAUGRUPPE (P-01, P-02, P-04):
• P-01 (DRW-P01-v2): Scheibe Ø220,0 x 3,0 mm EN AW-6061-T6, verschweißt mit Nabe Ø44,0 x 18,0 mm (X=-18..0). Fertigbearbeitung Nabenbohrung nach Schweißen: Ø17,02 H7 (+0,018/0), Querbohrung Ø6,0 H11 bei X = -9,0 mm.
• P-02 (DRW-P02-v2): Konischer Mantel X = 3,0..113,0 mm (L = 110,0 mm), R_außen(x) = 52,00 - (3,85/110)*(x-3,0), Wandstärke 2,0 mm, 3x axialer Umreifungskanal B = 16,0 mm x T = 6,0 mm bei 0°, 120°, 240° (REV-01).
• P-04 (DRW-P04-v2): Scheibe Ø220,0 x 3,0 mm mit Zentrierbund Ø30x1 mm, Bohrung Ø17,20 H8, 3x Schlitz 16x58 mm.

PROJEKT: NEox Compact 40M v2.0 | Fertigungszeichnung Flansche P-01/P-04 & Tro		
Werkstoff: EN AW-6061-T6	Masse: 0,856 kg (Satz)	Maßstab: 1:2,5
Zeichnungs-Nr.: DRW-ROT-003-v2	Toleranz: ISO 2768-mK	**Blatt 3/6**

P-07 Hauptspindel Ø17,00 k5 mit Schulter Ø20 und 3/8"-Antriebszapfen (L_ges = 248 mm)



P-09 Lagergehäuse (X=-84..-24)

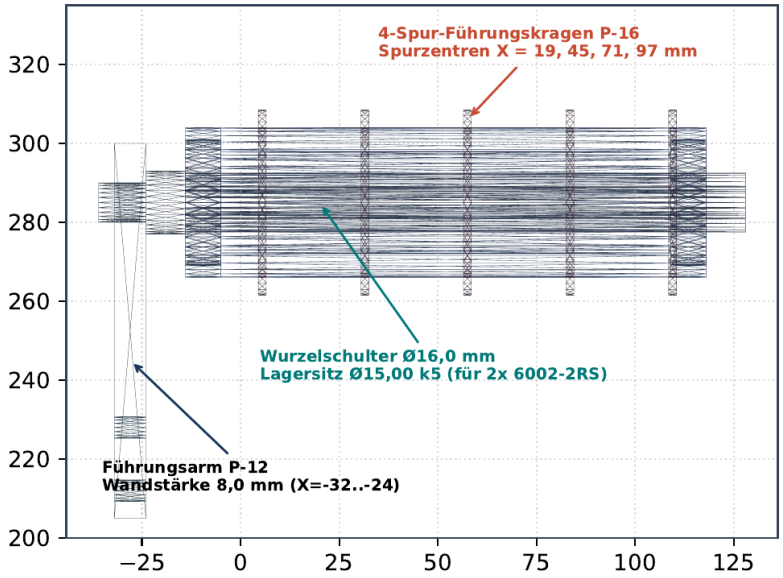


WELLEN- UND LAGERPASSUNGEN (REV-03 – KONFLIKTBEHEBUNG GEGENÜBER V1.0):

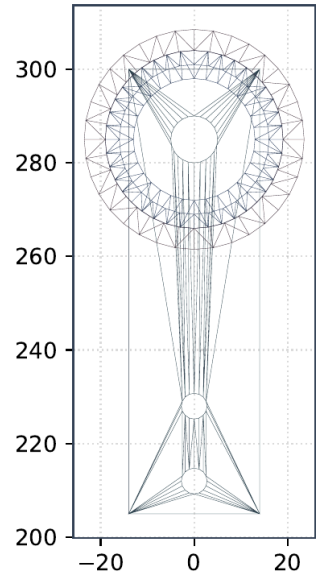
- Spindel P-07 (C45E +QT, geschliffen): Lagersitz- & Nabenbereich Ø17,00 k5 (+0,002 / +0,010 mm), Ra 0,8 µm. Behebt den kritischen v1.0-Fehler (dort Ø16,8 mm Welle mit 0,2 mm Spiel im 17-mm-Lager 6003-2RS).
- Lagergehäuse P-09 (EN AW-6061-T6): Beidseitige Lagersitzbohrung Ø35,00 H7 (+0,000 / +0,025 mm) x 10,0 mm tief bei X=[-84..-74] und X=[-34..-24] mit mittlerer Innenschulter Ø31,0 mm (X=[-74..-34]) zur axialen Festlegung.
- Innenring-Distanzhülse P-10 (EN AW-6061-T6): Ø22,0 / Ø17,10 x 40,00 mm (-0,02/+0,05 mm) stützt die Innenringe der beiden SKF 6003-2RS Lager beim Verspannen gegen die Wellenschulter Ø20,0 mm (X=-24 mm) ab.

PROJEKT: NEox Compact 40M v2.0 | Fertigungszeichnung Hauptspindel P-07 & Lag		
Werkstoff: C45E +QT / EN AW-6061-T6	Masse: 0,557 kg (Satz)	Maßstab: 1:2
Zeichnungs-Nr.: DRW-SPN-004-v2	Toleranz: ISO 2768-mK	**Blatt 4/6**

Baugruppe Bandführung (REV-02 & REV-06): P-12 bis P-16 bei Z = 285,0 mm

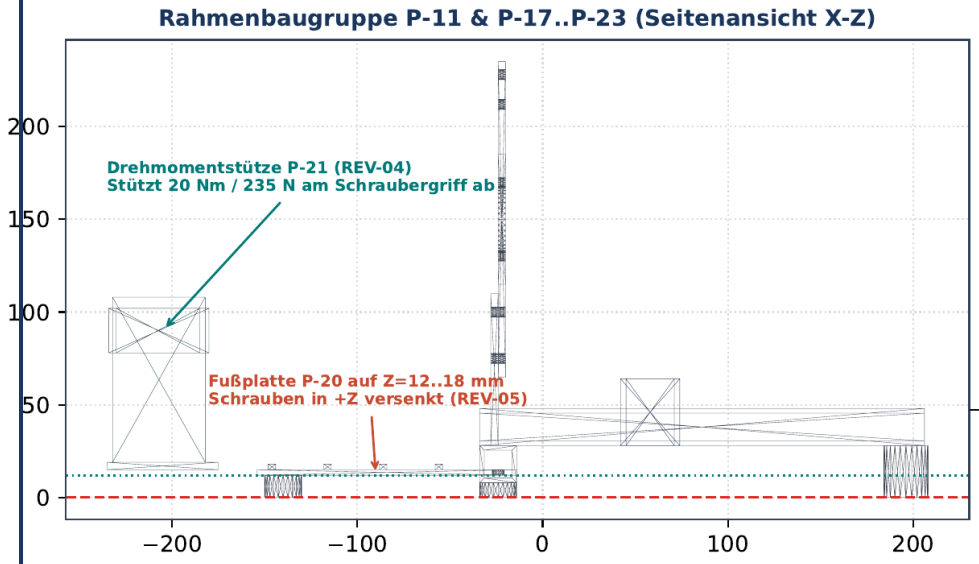


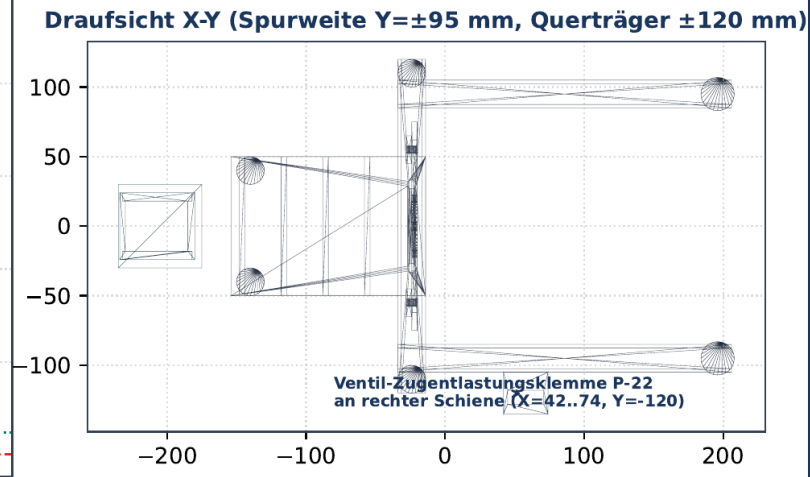
Seitenansicht Y-Z (Z=205..310)



KONSTRUKTIONSÄNDERUNG BANDFÜHRUNG (REV-02 & REV-06):
- Auflösung des Lagerkonflikts aus REV-02: Ein Rillenkugellager 6002-2RS (DIN 625) besitzt nominell d = 15,00 mm Innenbohrung, D = 32,00 mm Außendurchmesser und B = 9,00 mm Breite.
- Daher ist die Kragarm-Führungsachse P-13 an der Einspannwurzel (X = -24..-14 mm) mit Ø16,0 mm ausgeführt und am durchgehenden Lagersitz (X = -14..128 mm) exakt auf Ø15,00 k5 (+0,002/+0,010 mm) abgesetzt!
- Die Führungsrolle P-15 (POM-C) erhält Ø38,0 mm Außendurchmesser mit beidseitigen Lagersitzen Ø32,00 H7 x 9 mm.
- Biegesicherheit steigt von SF = 1,05 (v1.0: Ø8 mm) auf SF = 6,94 (v2.0: Ø15/16 mm), Durchbiegung sinkt auf 0,19 mm.

PROJEKT: NEox Compact 40M v2.0 | Fertigungszeichnung Bandführung P-12 bis P-1		
Werkstoff: C45E +QT / EN AW-6061-T6 / POM-C	Masse: 0,415 kg (Satz)	Maßstab: 1:2
Zeichnungs-Nr.: DRW-GDE-005-v2	Toleranz: ISO 2768-mK	**Blatt 5/6**





RAHMEN-, DREHMOMENT- UND BODENFREIHEITS-VALIDIERUNG (REV-04 & REV-05):

- Bodenfreiheit (REV-05): Fußplatte P-20 auf Z = 12,0..18,0 mm angehoben, gestützt durch 4x NBR-Elastomerpads P-23B (Z = 0..12 mm). Die M6-Rahmenschrauben F-17..F-20 werden von unten in +Z-Richtung in Flachsenkungen eingesetzt (Schraubenköpfe bei Z = 8..12 mm, Sicherungsmuttern oben bei Z = 28 bzw. 48 mm).
- Reaktionsmomentstütze P-21 (REV-04): Nimmt den Griff des INGCO CIDLI12206 (X = -230..-185, Y = ±18 mm) formschlüssig in einer NBR-gepolsterten Gabel auf (Auslegungslast 20 Nm -> 235,3 N Querlast, SF = 8,16).
- Zugentlastungsklemme P-22: Klemmt das Tropfband am Rahmen fest, sodass 0 N Zugkraft auf das Ventil wirken.

PROJEKT: NEox Compact 40M v2.0 | Fertigungszeichnung Rahmen, Drehmomentsti		
Werkstoff: EN AW-6061-T6 / NBR 70A	Masse: 0,968 kg (Satz)	Maßstab: 1:3
Zeichnungs-Nr.: DRW-FRM-006-v2	Toleranz: ISO 2768-mK	**Blatt 6/6**

8 Design evolution v1.0 → v2.0

Seven logged design changes, each checked by calculation and in the CAD geometry

Version 1.0 was a planning baseline. Reviewing it showed geometry, fit and strength conflicts that would have stopped the build. Revision 2.0 resolves every one of them and records the reason, the new geometry and the check result.

Rev.	Assembly	Implementation in revision 2.0	Result
REV-01	Rear flange P-01, drum P-02, front flange P-04	3 radial strap slots 16 mm wide, R = 45…103 mm at 0°, 120°, 240° with a closed outer ring R = 103…110 mm for stiffness; 3 axial drum channels 16×6 mm to pass the 3 straps P-03 under the inner tape layer	Released · flange bending stress 24.8 MPa, SF 9.68
REV-02	Guide arm P-12, axle P-13, bearings P-14, roller P-15	Arm wall raised to 8.0 mm (X = −32…−24, width 28 mm). Axle from C45E +QT: root Ø16.0, bearing seat Ø15.00 k5, hollow-drilled Ø8×120. 2× SKF 6002-2RS in a POM-C roller Ø38.0	Released · stress 408.5 → 62.0 MPa, SF 6.94
REV-03	Spindle P-07, hub P-01, housing P-09	Spindle Ø17.00 k5 with Ø20 shoulder; hub bore Ø17.02 H7; torque by M6×50 cross bolt F-21 (8.8) at X = −9.0 mm; housing with inner shoulder Ø31.0 and two seats Ø35.00 H7	Released · cross bolt SF 9.23, net shaft SF 5.80
REV-04	Drill torque support P-21	Aluminium support bolted to foot plate P-20 with an NBR-padded handle fork 45×36 mm; takes the full 20 Nm reaction (235.3 N at the handle) without blocking trigger or battery change	Released · bending 29.4 MPa, SF 8.16

Design changes continued

REV-05 to REV-07, and how the concept evolved

Rev.	Assembly	Implementation in revision 2.0	Result
REV-05	Frame P-18…P-20, feet P-23B	Foot plate P-20 raised to Z = 12.0…18.0 mm on 4× 12 mm elastomer pads P-23B; all M6 frame bolts F-17…F-20 inserted from below in +Z with recessed heads	Released · ≥ 12.0 mm ground clearance, no bolt protrudes
REV-06	4-lane guide P-16	Latching carriage with 5 guide collars (Ø47/Ø38) defines four 25 mm lanes at X = 19 / 45 / 71 / 97 mm so one continuous 40 m tape is traversed over 4 lanes of 10 m (R_coil 96.58 mm < 110 mm)	Released · matches the 110 mm drum
REV-07	Valve stem REF-03, clamp P-22	2 mm geometry gap at the valve handle stem closed (Z = 28…32 mm, R = 4 mm); frame-fixed tape strain-relief clamp P-22 added on the right floor rail	Released · valve fully load-isolated

Concept comparison

	Original reel	Compact concept	v2.0 tape winder
Frame	Square pipe + base	None: back plate + fold feet	Light aluminium hollow-profile frame on elastomer feet
Bearings	2 pillow blocks	1 flanged bearing	2× 6003-2RS spindle + 2× 6002-2RS guide roller
Guide	2 guide rollers	1 fairlead	1 roller with 4-lane carriage
Handling	Drill sticks out	Hex quick-connect	Hex stub + frame-fixed torque support
Envelope (concept stage)	28×28×24 cm ≈ 18.8 L	27×27×19 cm ≈ 14 L (−25%)	Flange Ø220, frame width 240, height 308.5 mm

The first two columns come from the concept note; the third shows the finished v2.0 geometry measured in the CAD model.

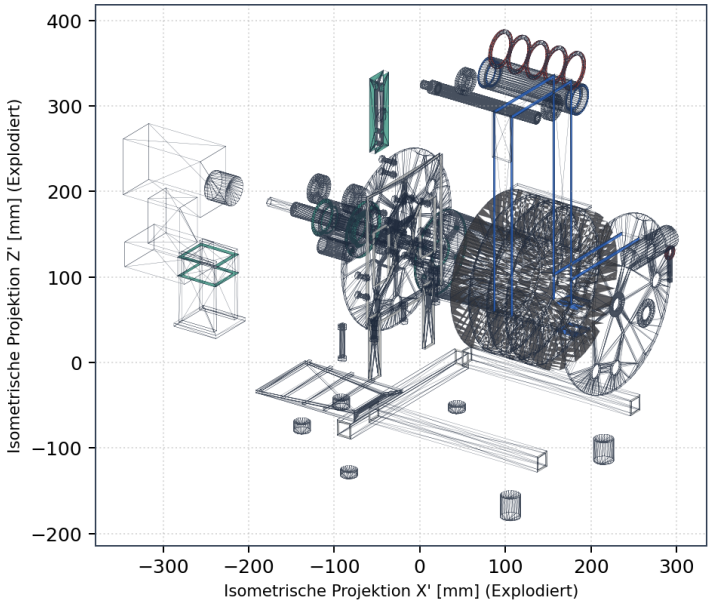


Figure 8 · Exploded isometric (detail of the report figure) showing the modular build: drive side left, drum and flanges right.

9 Conflict audit

Contradictions found in version 1.0 and how the numbers resolved them

ID	Conflict in v1.0	Resolution in v2.0
KONF-01 · Bearing 6003 vs. spindle	SKF 6003-2RS has d = 17.00, D = 35.00, B = 10.00 mm. v1.0 had a Ø16.8 shaft (0.2 mm play), hub Ø17.8/18.4 without a torque driver, housing R = 17.5 without shoulder, chassis bore only Ø24.	Spindle ground to Ø17.00 k5 (+2/+10 µm) with Ø20.0 shoulder (X = −24…−18) and spacer sleeve P-10 (Ø22/Ø17.1×40). Housing P-09 gets two Ø35.00 H7 seats (10 deep) and an inner shoulder Ø31.0 (X = −74…−34). Chassis bore opened to Ø36.0.
KONF-02 · Bearing 6002 vs. Ø16 axle	v1.0 axle Ø8 mm was too weak (SF 1.05) and the roller Ø24 too small. A 6002-2RS has a 15.00 mm bore (15×32×9), so it cannot sit on a Ø16 axle.	Axle P-13: root shoulder Ø16.0 (X = −24…−14), bearing seat Ø15.00 k5 (X = −14…128) for two 6002-2RS. Roller P-15 enlarged to Ø38.0 with two seats Ø32.00 H7.
KONF-03 · Cone formula vs. 2°	R(x) = 52.0 − (3.85/110)(x − 3.0) gives R(113) = 48.15 mm and ΔR = 3.85 mm, which is α = arctan(3.85/110) = 2.0045°. Exactly 2.0000° would give ΔR = 3.8413 mm.	The linear definition (52.00 → 48.15 mm) is the master geometry. The 0.0087 mm deviation is 20× inside ISO 2768-m, and 48.15 mm keeps the 48.0 mm minimum bend radius.
KONF-04 · One tape vs. 4 rings	v1.0 modelled four separate tape rings although the machine winds one continuous tape. A single 25 mm stack (N = 1) would reach R_coil = 172.58 mm, beyond R_flange = 110 mm.	One tape traversed by carriage P-16 over 4 connected lanes of 10 m (about 21.7 turns per lane). R_coil = 96.58 mm, 13.42 mm flange reserve.
KONF-05 · Fastener ground clearance	v1.0 M6 nuts at Z = 1…8 mm touched the ground and collided with foot plate P-20 (Z = 6…8).	Foot plate at Z = 12…18 on 12 mm NBR pads; bolts F-17…F-20 turned to +Z (heads Z = 8…12 recessed, nuts on top at Z = 28/48).

10 Geometry audit

Values changed between the planning baseline and the released geometry (from the calculation workbook, sheet “Geometrie”)

ID	Feature	v1.0	v2.0	Δ	Assessment
GEO-01	Drum width, 4 lanes × 25 mm	110	110	0	100 mm tape + 3 mm steps + 7 mm margin
GEO-02	Drum outer radius rear (X = 3)	52.00	52.00	0	Unchanged
GEO-03	Drum outer radius front (X = 113)	48.15	48.15	0	Above 48.0 mm minimum bend radius
GEO-04	Radial taper ΔR over 110 mm	3.85	3.85	0	Linear master equation governs
GEO-05	Cone angle from ΔR = 3.85	2.000°	2.005°	+0.005°	2.0° is the nominal value
GEO-07	Spindle bearing seat Ø (6003, d=17)	16.8	17.00	+0.20	Conflict solved: v1.0 had 0.2 mm play
GEO-08	Rear hub bore	17.8	17.02	−0.78	Conflict solved: H7 fit + cross pin
GEO-09	Housing bearing seat (D=35)	35	35	0	Inner shoulder Ø31 added
GEO-10	Chassis through-bore	24	36	+12	Opened for assembly clearance
GEO-11	Guide arm wall in X	4	8	+4	Doubled for cantilever stiffness
GEO-12	Guide axle at root / bearing	8	16 / 15.00 k5	+7	Fits 6002-2RS bore 15.00
GEO-13	Guide roller outer Ø	24	38	+14	v1.0 was smaller than bearing D = 32
GEO-14	Foot plate underside height	6	12	+6	Raised to Z = 12 mm
GEO-15	Minimum fastener clearance	1	12	+11	No protruding nuts underneath
GEO-16	Valve stem Z extent	30	28	−2	2 mm gap closed (Z = 28…32)

Dimensions in mm unless marked. “Δ” is v2.0 minus v1.0.

Each row ties to a model dimension, so a reviewer can open the CAD, measure the feature and compare with this table. The audit is the reason the spindle, bearings and housing now agree to within standard ISO 286 fits.

11 Drum geometry and capacity

Why four lanes and a 2° cone are required

Drum longitudinal section (X–R): 2° cone, 4 tape lanes, flanges Ø220

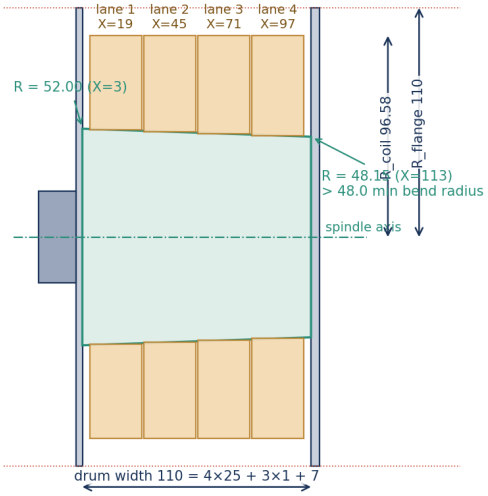


Figure 9 · Longitudinal section through the drum drawn from the master geometry. Lane centres sit at X = 19, 45, 71, 97 mm; R_coil = 96.58 mm leaves 13.42 mm to the flange edge.

Governing equations

Quantity	Formula	Value
Drum width	W = N·W_tape + (N − 1)·s_lane + 2·c_flange = 4·25 + 3·1 + 7	110.0 mm
Mean core radius	R_core,mean = ½·(52.00 + 48.15)	50.075 mm
Outer coil radius	R_coil = √(R_core,mean² + L_tape·t_eff / (π·N·η)) = √(50.075² + 40 000·1.5 / (π·4·0.70))	96.58 mm (Ø193.17)
Single-lane check	N = 1: √(50.075² + 60 000 / (π·0.70))	172.58 mm (62.58 mm over flange)
Spindle turns	N_total = L / (2π·R_m), R_m = 73.33 mm, C_m = 460.74 mm/turn	86.82 turns (≈ 21.7 per lane)

Inputs: tape length 40 000 mm, lay-flat width 25.0 mm (24.5…25.5), effective double-wall thickness including emitter t_eff = 1.5 mm (assumption, to be measured), packing factor η = 0.70 (conservative assumption), lane gap 1.0 mm, flange clearance 3.5 mm each side.

12 Capacity sensitivity

What happens if the tape is thicker, thinner or packed differently

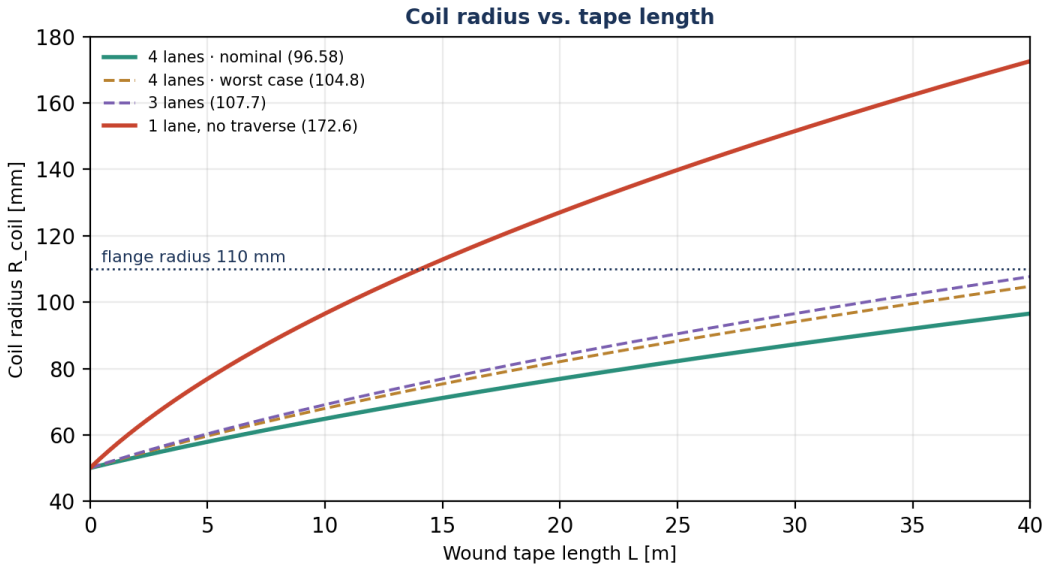


Figure 10 · R_coil versus wound length for four cases. Only the traversed 4-lane layout stays inside the 110 mm flange radius.

ID	Winding mode (one 40 m tape)	N	t_eff	η	R_coil	D_coil	Flange reserve
CAP-01	4 lanes, nominal, R_core = 50.0	4	1.50	0.70	96.54	193.1	13.46
CAP-02	4 lanes, exact mean cone R_core = 50.075	4	1.50	0.70	96.58	193.2	13.42
CAP-03	4 lanes, dense packing	4	1.40	0.78	90.67	181.3	19.33
CAP-04	4 lanes, worst case	4	1.65	0.62	104.8	209.6	5.22
CAP-05	3 lanes (one lane unused)	3	1.50	0.70	107.7	215.4	2.29
CAP-06	1 lane, no traverse	1	1.50	0.70	172.6	345.2	−62.6
CAP-07	1 lane, thin tape	1	1.20	0.80	147.0	294.0	−37.0

Radii and diameters in mm. Even in the worst-case packing (CAP-04) the coil clears the flange by 5.2 mm, which is why measuring the real tape thickness before production is listed as a validation task.

13 Drive, speed and winding time

Why a speed-reducing gearbox between drill and spindle is mandatory

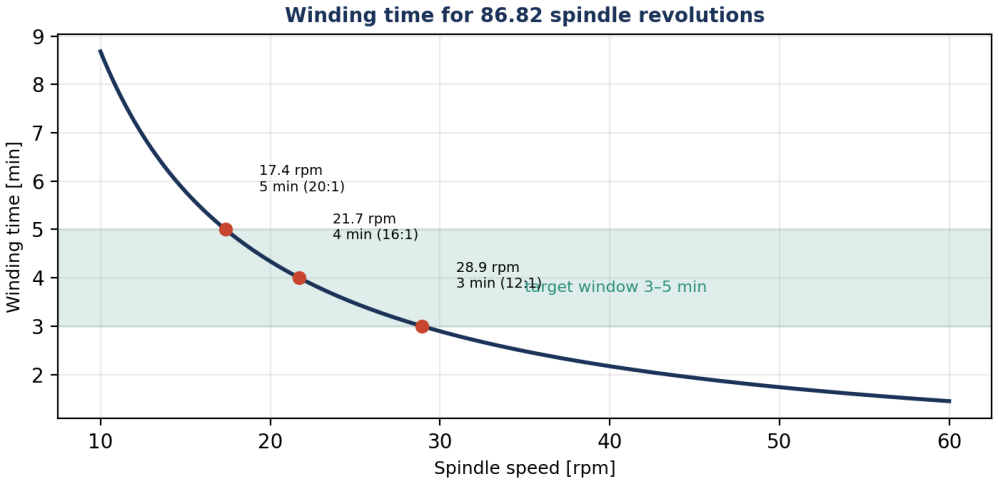


Figure 11 · Winding time is 86.82 turns divided by spindle speed. The 3–5 minute target needs 17.4–28.9 rpm.

ID	Drive configuration	n [rpm]	t [min]	v_start [m/s]	v_end [m/s]	Assessment
RPM-01	Target minimum, gear 12:1	28.94	3.00	0.152	0.293	Released
RPM-02	Nominal design point, gear 16:1	21.70	4.00	0.114	0.220	Released · about 60 s per 10 m lane
RPM-03	Target maximum, gear 20:1	17.36	5.00	0.091	0.176	Released
RPM-04	Direct, 1st gear, light trigger	60	1.45	0.315	0.607	Warning: too fast for hand guidance
RPM-05	Direct, 1st gear, 350 rpm	350	0.25	1.835	3.54	Locked out
RPM-06	Direct, 1st gear, 400 rpm full speed	400	0.22	2.098	4.046	Locked out · gearbox required
RPM-07	Direct, 2nd gear, 1500 rpm	1500	0.058	7.866	15.17	Locked out · block 2nd gear

RISK-02. The INGCO CIDLI12206 runs 0–400 rpm in 1st gear and 0–1500 rpm in 2nd. Driven 1:1 it would wind the whole tape in 13–15 seconds at up to 3.5 m/s. A compact planetary reduction of 12:1 to 16:1 between chuck and spindle is therefore required, with the clutch set to stage 6–8 (about 8 Nm at the spindle) so that tape pull stays near 83–120 N, well below the tape breaking load of about 280 N.

14 Torque and reaction forces

Load cases from normal winding to the worst-case jam at the drill’s maximum torque

ID	Load case	F_pull [N]	Lever R [mm]	M_spindle [Nm]	F_grip (r = 85 mm) [N]	Limit	Result
TRQ-01	Start of winding at core, nominal dry	35	52	1.82	21.4	8.0 Nm (clutch stage 6)	OK
TRQ-02	End of winding at outer radius, dry	35	96.58	3.38	39.8	8.0 Nm	OK
TRQ-03	End of winding, wet / dirty tape	65	96.58	6.28	73.9	8.0 Nm	OK
TRQ-04	Jam, clutch stage 6	120	96.58	11.59	136.4	12.0 Nm (clutch trips)	OK · tape break load > 280 N not reached
TRQ-05	Maximum catalogue stall torque (REV-04)	207.1	96.58	20.0	235.3	20.0 Nm (structural design load)	OK · support P-21 holds 235.3 N

Reading the table

- Normal winding needs only 1.8–3.4 Nm. Even wet, dirty tape (65 N) stays below the clutch setting.
- A jam is limited by the clutch; if the clutch fails, the stall torque of 20 Nm is the design load for every structural part, and the numbers on the next page show every safety factor against it.
- The reaction force at the drill handle (235.3 N) is carried by the frame-fixed support P-21, so the operator never has to hold it.
- Tape pull values of 35 N and 120 N are **assumptions** from bench reasoning and must be confirmed by a pull-scale test on dry field soil.

15 Strength verification

Analytical checks for the worst case: 120 N tape pull at the outermost lane and 20 Nm stall torque

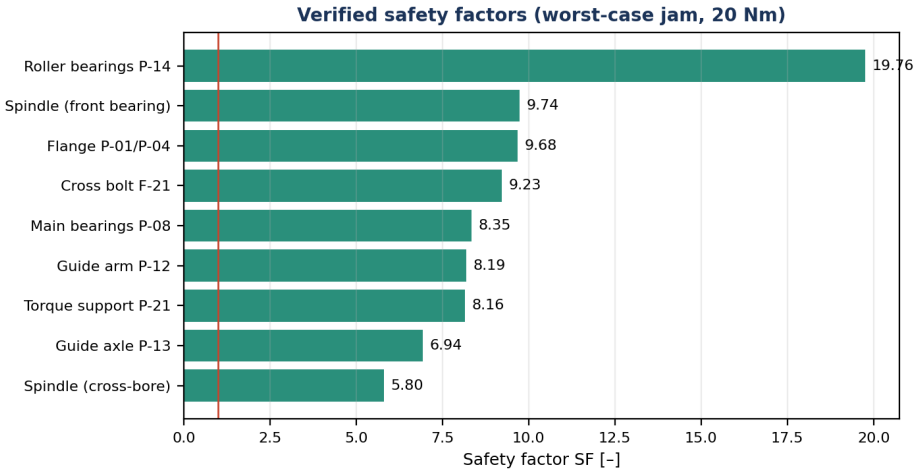


Figure 12 · Safety factors from the engineering report. All values are far above 1.0.

Component · location	Load	Stress / force	Allowable	SF	Note
Guide axle P-13, seat Ø15 / root Ø16	Bending from F_res = √2·120 = 169.7 N (M_b = 20.53 Nm)	62.0 MPa (Ø15) · 51.1 MPa (Ø16)	430 MPa	6.94	Deflection 0.19 mm instead of 2.37 mm in v1.0
Main spindle P-07, front bearing X = −29	Bending 12.39 Nm + torsion 20.0 Nm (von Mises)	44.2 MPa	430 MPa	9.74	Large fatigue reserve
Main spindle P-07, cross bore X = −9	Net section Ø17 with Ø6 bore, 20.0 Nm	74.1 MPa	430 MPa	5.80	Notch effect uncritical at 22 rpm
Cross bolt F-21 (M6×50, 8.8)	Double shear, F = T/d = 1 176.5 N	41.6 MPa	384 MPa	9.23	Play-free form-fit drive
Main bearings P-08 (6003-2RS)	Radial load at front bearing 389.4 N	389.4 N	C0 = 3 250 N	8.35	L10 > 4 300 million revolutions
Roller bearings P-14 (6002-2RS)	Radial load on outer lane 144.3 N	144.3 N	C0 = 2 850 N	19.76	Two-bearing support, no tilting
Guide arm P-12 (6061-T6)	M_b = 120 N · 73 mm = 8.76 Nm	29.3 MPa	240 MPa	8.19	Wall doubled from 4 to 8 mm
Torque support P-21	F_grip = 20 Nm / 0.085 m = 235.3 N at Z = 90 mm	29.4 MPa	240 MPa	8.16	2 cheeks 35 × 4.0 mm
Flanges P-01 / P-04	Bending	24.8 MPa	240 MPa	9.68	With closed outer ring

16 Bearings, shafts and fits

Every rolling bearing sits on a standard ISO 286 fit; nothing relies on play

Joint	Parts	Type and axis	Fit / nominal	Fastener	Acceptance check
JNT-01	P-07 spindle ↔ P-08A/B (6003-2RS)	Rolling-bearing shaft fit (X)	Ø17.00 k5 (+2/+10 µm)	P-10 spacer	Micrometer Ø17.002…17.010; no play
JNT-02	P-08A/B ↔ P-09 housing	Stepped bearing seat (X)	Ø35.00 H7 (+0/+25 µm)	Shoulder Ø31.0	Go/no-go plug gauge; press-in depth 10.0 mm
JNT-03	P-09 ↔ P-11 chassis	Face flange bolting (X = −24)	4× Ø5.5 on PCD R = 28	F-01…04 (M5×16)	Torque 5.5 Nm; spindle squareness
JNT-04	P-01 hub ↔ P-07 spindle	Form-fit pin and hub (X = −9)	Ø17.02 H7 / Ø17.00 k5 + Ø6	F-21 (M6×50 8.8)	Stall-torque test 20 Nm without slip
JNT-05	P-01 ↔ P-02 drum	Axial flange bolting (X = 3)	6× Ø5.5 on PCD R = 38	F-11…16 (M5×12)	Flange runout < 0.15 mm; slots align with channels
JNT-06	P-04 front flange ↔ P-02 and P-07	Quick-release push fit (+X)	Ø17.20 H8 on Ø17.00 k5	P-05 + pin P-06	Tool-free removal in < 5 s
JNT-07	P-12 arm ↔ P-11 chassis	Double-shear bolting (X = −24)	2× Ø5.5 at Z = 212, 228	F-09, F-10 (M5×20)	5.5 Nm; axle parallel to spindle
JNT-08	P-13 axle ↔ P-12 and P-14	Shoulder bolt + bearing seat (Z = 285)	Root Ø16 / seat Ø15.00 k5	F-22 (M10×1.5)	35 Nm; roller turns freely
JNT-09	P-18 ↔ P-19L/R and P-20	Ground-clearance bolting (+Z)	Heads recessed Z = 8…12	F-17…20 (M6)	Clearance gauge ≥ 12.0 mm
JNT-10	P-21 ↔ P-20 and drill	Form-fit handle fork	Fork 45×36 mm with NBR	F-23…26 (M5×16)	20 Nm reaction test; battery change free
JNT-11	P-22 ↔ P-19R and tape	Clamping strain relief	Channel 26 × 4 mm	F-27, F-28 (M5×16)	120 N pull test → 0 N at valve

Joint register JNT-01 … JNT-11 from the engineering report.

17 Mass budget

Material-based CAD volumes give 3.097 kg dry, 1.903 kg under the 5.000 kg limit

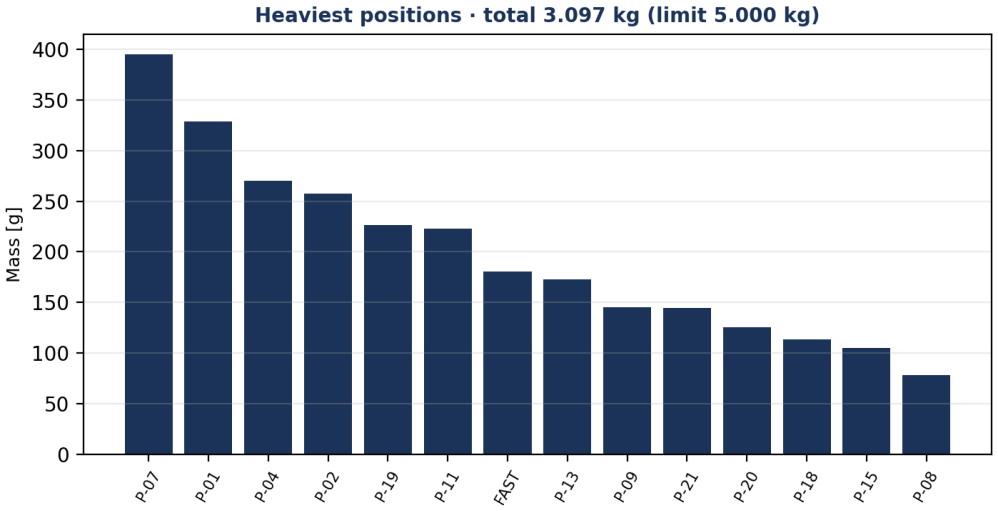


Figure 13 · The 14 heaviest positions (grams).

ID	Part	Qty	Material	Mass [kg]
P-01	Rear flange with welded hub	1	EN AW-6061-T6	0.3285
P-02	Conical drum with 3 strap channels	1	EN AW-6061-T6	0.2577
P-03	Hook-and-loop strapping bands	3	PP/PA webbing	0.0233
P-04	Front removable guard flange	1	EN AW-6061-T6	0.2701
P-05	Axial retaining washer	1	C45E galvanised	0.0106
P-06	Locking pin with pull ring	1	Galvanised steel	0.0091
P-07	Main spindle with 3/8" drive stub	1	C45E +QT	0.3954
P-08	Deep-groove ball bearing 6003-2RS	2	100Cr6 (SKF)	0.0779
P-09	Stepped bearing housing	1	EN AW-6061-T6	0.1452
P-10	Bearing spacer sleeve	1	EN AW-6061-T6	0.0163
P-11	Main chassis plate	1	EN AW-6061-T6	0.2230
P-12	Reinforced guide arm (8 mm)	1	EN AW-6061-T6	0.0547
P-13	Cantilever guide axle	1	C45E +QT	0.1728
P-14	Deep-groove ball bearing 6002-2RS	2	100Cr6 (SKF)	0.0600
P-15	Tape guide roller Ø38	1	POM-C	0.1053
P-16	4-lane tape guide carriage	1	POM-C	0.0217
P-17	Leg strips, left & right	2	EN AW-6061-T6	0.0344
P-18	Cross member, hollow 20×20	1	EN AW-6061-T6	0.1134
P-19	Floor rails, left & right	2	EN AW-6061-T6	0.2268
P-20	Foot plate (Z = 12…18)	1	EN AW-6061-T6	0.1258
P-21	Drill torque support	1	EN AW-6061-T6 + NBR	0.1447
P-22	Valve strain-relief clamp	1	EN AW-6061-T6 + POM	0.0443
P-23	Elastomer feet & pads (A: 2 front, B: 4 rear)	6	NBR 70 Shore A	0.0552
FAST	Standard fastener set (F-01…F-34)	34	Steel 8.8 galvanised / A2	0.1810
Total	**Machine, dry (no drill, battery or tape)**		**limit ≤ 5.000 kg**	**3.097**

18 Part catalogue

Every part with its shaded views (side, front, top, isometric) rendered from the STL files, and the data from the bill of materials

P-01 Rear flange with welded hub

Hinterer Flansch mit Nabe

Side X-Z | Front Y-Z | Top X-Y | Isometric

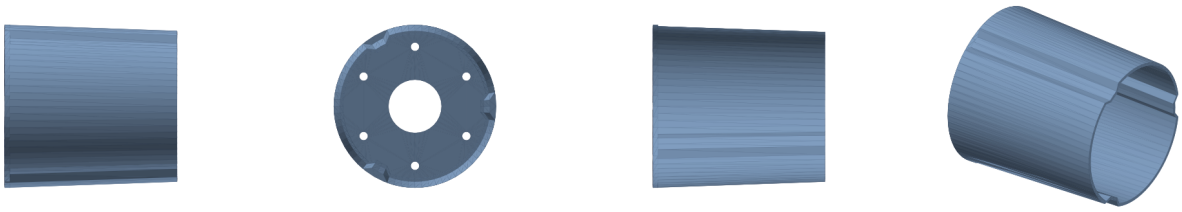


Qty / material	1 · EN AW-6061-T6	**Main dimensions**	Ø220×3 disc + hub Ø44/Ø17×18
Process	Laser cutting + turning + TIG welding	**Bores & fits**	Hub bore Ø17.02 H7; cross hole Ø6.0 H11; 6× Ø5.5 on PCD R=38; 3 radial slots 16×58
Position [mm]	X=[−18…3], Y=0, Z=150	**Mounted with**	P-02, P-07 · F-11…16, F-21
Mass (total)	0.3285 kg		

P-02 Conical drum with 3 strap channels

Konische Trommel mit Umreifungskanälen

Side X-Z | Front Y-Z | Top X-Y | Isometric

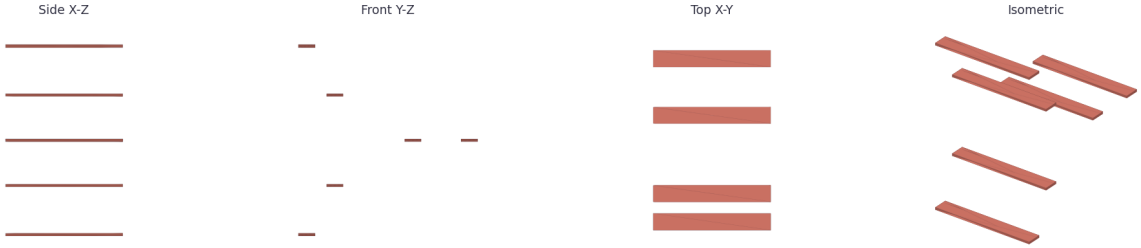


Qty / material	1 · EN AW-6061-T6	**Main dimensions**	Ø104/Ø96.3 × 110, wall 2.0
Process	Flow forming / CNC turning + milling	**Bores & fits**	6× Ø5.5 on PCD R=38; 3 axial channels 16×6 at 0°/120°/240°
Position [mm]	X=[3…113], Y=0, Z=150	**Mounted with**	P-01, P-03, P-04 · F-11…16
Mass (total)	0.2577 kg		

P-03 Hook-and-loop strapping bands

Klett-Umreifungsbänder

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	3 · PP/PA webbing	**Main dimensions**	15 × 380 × 1.5
Process	Purchased, made up	**Bores & fits**	Metal/PA buckle loop
Position [mm]	X=[5…111] at 0°/120°/240°	**Mounted with**	P-02, REF-01
Mass (total)	0.0233 kg		

P-04 Front removable guard flange

Vorderer Schutzflansch

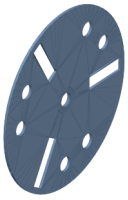
Side X-Z

Front Y-Z



Top X-Y

Isometric



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	Ø220×3 + collar Ø30×1
Process	Laser cutting + CNC centring collar	**Bores & fits**	Centring bore Ø17.20 H8; 6× Ø18; 3 slots 16×58
Position [mm]	X=[113…117], Y=0, Z=150	**Mounted with**	P-02, P-05, P-07
Mass (total)	0.2701 kg		

P-05 Axial retaining washer

Axiale Haltescheibe

Side X-Z



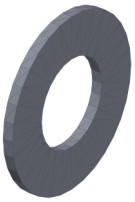
Front Y-Z



Top X-Y



Isometric



Qty / material	1 · C45E galvanised	**Main dimensions**	Ø34 / Ø17.2 × 2.0
Process	Turning / stamping (DIN 988)	**Bores & fits**	Bore Ø17.20 (+0.20)
Position [mm]	X=[117…119], Y=0, Z=150	**Mounted with**	P-04, P-06, P-07
Mass (total)	0.0106 kg		

P-06 Locking pin with pull ring

Sicherungspin (Klappstecker)

Side X-Z



Front Y-Z



Top X-Y



Isometric



Qty / material	1 · Galvanised steel	**Main dimensions**	Shank Ø6.0 × 30, ring Ø17
Process	Standard part DIN 11023	**Bores & fits**	Mates with Ø6.2 bore in P-07
Position [mm]	X=122, Y=0, Z=[135…175]	**Mounted with**	P-05, P-07
Mass (total)	0.0091 kg		

P-07 Main spindle with 3/8" drive stub

Hauptspindel mit 3/8"-Antriebszapfen

Side X-Z | Front Y-Z | Top X-Y | Isometric

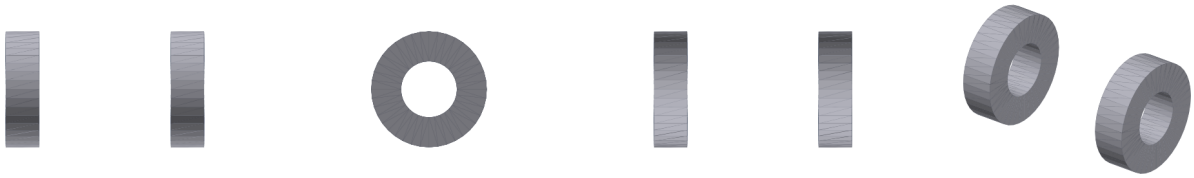


Qty / material	1 · C45E +QT	**Main dimensions**	Ø17.00 k5 × 212 + hex AF 9.5 × 36
Process	CNC turning + milling + grinding	**Bores & fits**	Bearing seat Ø17.00 k5; shoulder Ø20; cross hole Ø6.0 (X=−9); pin hole Ø6.2 (X=122)
Position [mm]	X=[−120…128], Y=0, Z=150	**Mounted with**	P-01, P-04…06, P-08, P-10 · F-21
Mass (total)	0.3954 kg		

P-08 Deep-groove ball bearing 6003-2RS

Rillenkugellager 6003-2RS

Side X-Z | Front Y-Z | Top X-Y | Isometric

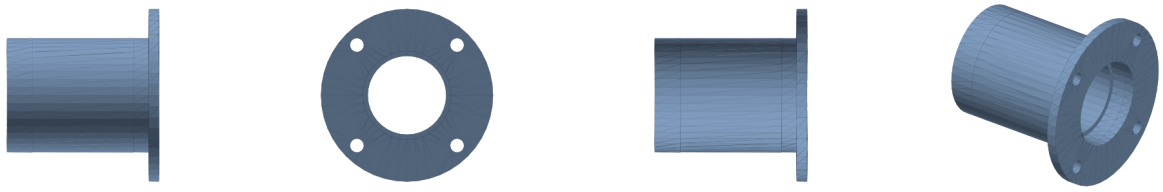


Qty / material	2 · 100Cr6 (SKF)	**Main dimensions**	d=17.00 × D=35.00 × B=10.00
Process	Standard part, DIN 625-1	**Bores & fits**	Inner ring on Ø17 k5; outer ring in Ø35 H7
Position [mm]	X=[−84…−74] & [−34…−24], Z=150	**Mounted with**	P-07, P-09, P-10
Mass (total)	0.0779 kg		

P-09 Stepped bearing housing

Abgesetztes Lagergehäuse

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	Tube Ø45×56, flange Ø68×4
Process	CNC turning + drilling	**Bores & fits**	2× bearing seat Ø35.00 H7 (10 deep); inner shoulder Ø31; 4× Ø5.5 on PCD R=28
Position [mm]	X=[−84…−24], Y=0, Z=150	**Mounted with**	P-08, P-11 · F-01…04
Mass (total)	0.1452 kg		

P-10 Bearing spacer sleeve

Lager-Distanzhülse

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	Ø22.0 / Ø17.1 × 40.0
Process	Turned from tube	**Bores & fits**	Bore Ø17.10 (+0.10); runout 0.02
Position [mm]	X=[−74…−34], Y=0, Z=150	**Mounted with**	P-07, P-08A/B
Mass (total)	0.0163 kg		

P-11 Main chassis plate

Haupt-Chassisplatte

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	150 × 170 × 4.0
Process	Laser cutting + deburring	**Bores & fits**	Central Ø36; 4× Ø5.5 (PCD 28); 4× Ø5.5 (legs); 2× Ø5.5 (arm); 2 lightening windows 30×70
Position [mm]	X=[−24…−20], Y=[−75…75], Z=[65…235]	**Mounted with**	P-09, P-12, P-17 · F-01…10
Mass (total)	0.2230 kg		

P-12 Reinforced guide arm (8 mm)

Verstärkter Führungsarm

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	28 × 95 × 8.0
Process	CNC milling / waterjet	**Bores & fits**	2× Ø5.5 (Z=212, 228); 1× Ø10.2 (Z=285)
Position [mm]	X=[−32…−24], Y=[−14…14], Z=[205…300]	**Mounted with**	P-11, P-13 · F-09/10, F-22
Mass (total)	0.0547 kg		

P-13 Cantilever guide axle

Kragarm-Führungsachse

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	1 · C45E +QT	**Main dimensions**	M10×12 + Ø16×10 + Ø15.00 k5 × 142
Process	CNC turning + deep-hole drilling	**Bores & fits**	Bearing seat Ø15.00 k5; hollow bore Ø8×120; thread M10×1.5
Position [mm]	X=[−36…128], Y=0, Z=285	**Mounted with**	P-12, P-14 · F-22
Mass (total)	0.1728 kg		

P-14 Deep-groove ball bearing 6002-2RS

Rillenkugellager 6002-2RS

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	2 · 100Cr6 (SKF)	**Main dimensions**	d=15.00 × D=32.00 × B=9.00
Process	Standard part, DIN 625-1	**Bores & fits**	Inner ring on Ø15 k5; outer ring in Ø32 H7
Position [mm]	X=[−14…−5] & [109…118], Z=285	**Mounted with**	P-13, P-15
Mass (total)	0.0600 kg		

P-15 Tape guide roller Ø38

Band-Führungsrolle

Side X-Z | Front Y-Z | Top X-Y | Isometric



Qty / material	1 · POM-C	**Main dimensions**	Ø38.0 / Ø26.0 × 132.0
Process	CNC turning	**Bores & fits**	2× bearing seat Ø32.00 H7 × 9.0 deep
Position [mm]	X=[−14…118], Y=0, Z=285	**Mounted with**	P-14, P-16
Mass (total)	0.1053 kg		

P-16 4-lane tape guide carriage

4-Spur-Bandführungsschlitten



Qty / material	1 · POM-C	**Main dimensions**	5 guide collars Ø47/Ø38 × 2.0
Process	CNC milling / turning	**Bores & fits**	Lane centres X = 19, 45, 71, 97 (lane width 26)
Position [mm]	X=[4.5…110.5], Y=0, Z=285	**Mounted with**	P-15, REF-01
Mass (total)	0.0217 kg		

P-17 Leg strips, left & right

Beinleisten Links & Rechts



Qty / material	2 · EN AW-6061-T6	**Main dimensions**	20 × 82 × 4.0
Process	Laser cutting + welding to P-18	**Bores & fits**	Each 2× Ø5.5 at Z=75 and Z=100
Position [mm]	X=[−28…−24], Y=±[45…65], Z=[28…110]	**Mounted with**	P-11, P-18 · F-05…08
Mass (total)	0.0344 kg		

P-18 Cross member, hollow 20×20

Querträger Hohlprofil 20x20



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	20 × 20 × 2.5 × 240
Process	Sawing + drilling + counterboring	**Bores & fits**	2× Ø6.5 (Y=±95), 2× Ø6.5 (Y=±30); flat counterbore on underside Z=8…12
Position [mm]	X=[−34…−14], Y=[−120…120], Z=[8…28]	**Mounted with**	P-17, P-19, P-20, P-23B · F-17…20
Mass (total)	0.1134 kg		

P-19 Floor rails, left & right

Bodenschienen Links & Rechts



Qty / material	2 · EN AW-6061-T6	**Main dimensions**	20 × 20 × 2.5 × 240
Process	Sawing + drilling	**Bores & fits**	Each 1× Ø6.5 (X=−24), 1× Ø5.5 (X=196), 2× Ø5.5 (strain relief)
Position [mm]	X=[−34…206], Y=±[85…105], Z=[28…48]	**Mounted with**	P-18, P-22, P-23A · F-17/18, F-27…30
Mass (total)	0.2268 kg		

P-20 Foot plate (Z = 12…18)

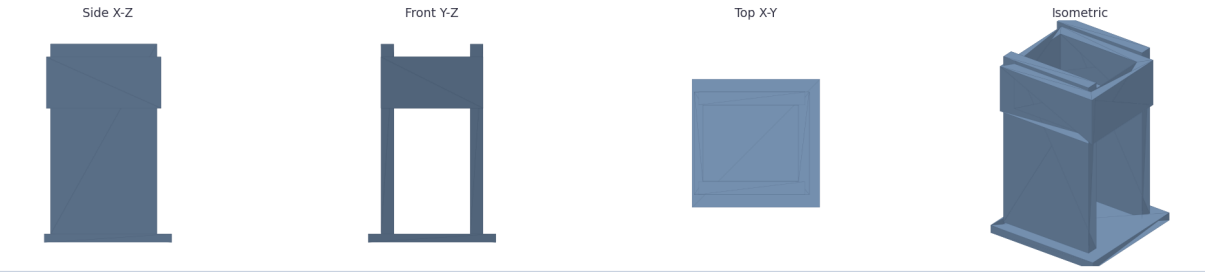
Fußplatte



Qty / material	1 · EN AW-6061-T6	**Main dimensions**	140 × 100 × 3.0 + 4 ribs 4×3
Process	Milling / extruded profile	**Bores & fits**	2× Ø6.5 (X=−24, Y=±30); 4× Ø5.5 (torque support); 2× Ø5.5 (pads)
Position [mm]	X=[−154…−14], Y=[−50…50], Z=[12…18]	**Mounted with**	P-18, P-21, P-23B · F-19…26
Mass (total)	0.1258 kg		

P-21 Drill torque support

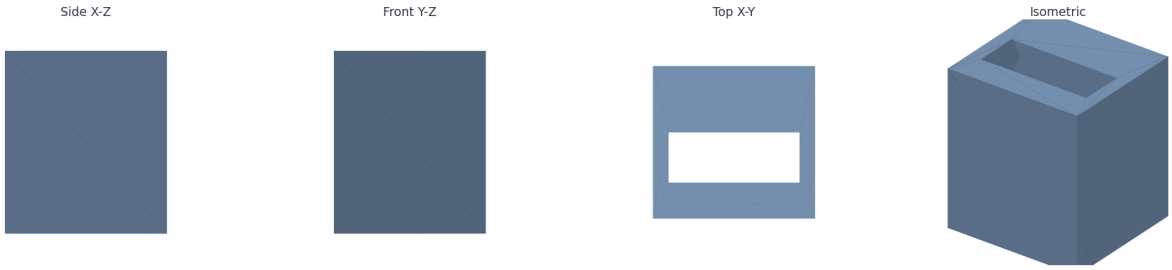
Bohrschrauber-Drehmomentstütze



Qty / material	1 · EN AW-6061-T6 + NBR	**Main dimensions**	60 × 60 × 93; handle fork 45 × 36
Process	Bending + milling + NBR lining	**Bores & fits**	4× Ø5.5 base flange; 2× M5 clamp screws
Position [mm]	X=[−235…−175], Y=[−30…30], Z=[15…108]	**Mounted with**	P-20, REF-02 · F-23…26
Mass (total)	0.1447 kg		

P-22 Valve strain-relief clamp

Ventil-Zugentlastungsklemme



Qty / material	1 · EN AW-6061-T6 + POM	**Main dimensions**	32 × 30 × 36; tape channel 26 × 4
Process	CNC milling	**Bores & fits**	2× Ø5.5 mounting to right floor rail; quick clamp
Position [mm]	X=[42…74], Y=[−135…−105], Z=[28…64]	**Mounted with**	P-19R, REF-03 · F-27/28
Mass (total)	0.0443 kg		

P-23 Elastomer feet & pads (A: 2 front, B: 4 rear)

Elastomer-Standfüße & Pads



Qty / material	6 · NBR 70 Shore A	**Main dimensions**	A: Ø24/Ø6.0 × 28 · B: Ø24/Ø5.5 × 12
Process	Purchased / moulded	**Bores & fits**	Through-bore Ø5.5, counterbored
Position [mm]	A: X=196, Y=±95, Z=[0…28] · B: X=−24 (Y=±110) & X=−140 (Y=±40), Z=[0…12]	**Mounted with**	P-18, P-19, P-20 · F-29…34
Mass (total)	0.0552 kg		

FAST Standard fastener set (F-01…F-34)

Normbefestigersatz

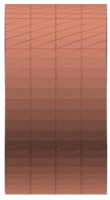


Qty / material	34 · Steel 8.8 galvanised / A2	**Main dimensions**	24× M5, 5× M6, 1× M10×1.5 incl. washers & nuts
Process	Purchased standard parts	**Bores & fits**	All bolt heads Z ≥ 12 mm above ground (REV-05)
Position [mm]	See fastener register	**Mounted with**	All frame and rotor joints
Mass (total)	0.1810 kg		

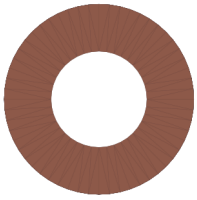
REF-01 40 m ALLWIN GOLD drip-tape roll (reference)

40m ALLWIN GOLD Tropfband-Rolle

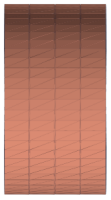
Side X-Z



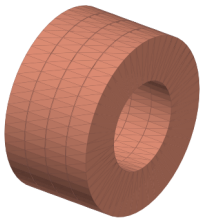
Front Y-Z



Top X-Y



Isometric

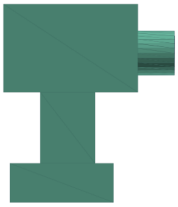


Qty / material	1 · Reference geometry	**Main dimensions**	Coil Ø193.2, tape 25 mm lay-flat
Process	—	**Bores & fits**	Envelope for collision checks
Position [mm]	On drum P-02	**Mounted with**	—
Mass (total)	—		

REF-02 INGCO CIDLI12206 12 V drill (reference)

INGCO CIDLI12206 12V Bohrschrauber

Side X-Z



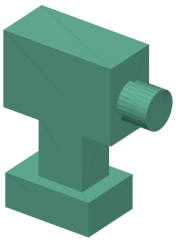
Front Y-Z



Top X-Y



Isometric



Qty / material	1 · Reference geometry	**Main dimensions**	Handle envelope X = −230…−185
Process	—	**Bores & fits**	Gripped in P-21 fork
Position [mm]	X ≤ −175	**Mounted with**	—
Mass (total)	—		

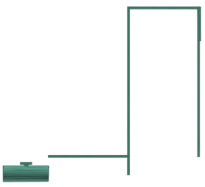
REF-03 Stationary valve connection & tape path (reference)

Stationärer Ventilanschluss & Bandpfad

Side X-Z



Front Y-Z



Top X-Y



Isometric



Qty / material	1 · Reference geometry	**Main dimensions**	Valve at Y = −262…−340
Process	—	**Bores & fits**	Valve stem gap closed (REV-07)
Position [mm]	X=58, Y=−262, Z=16	**Mounted with**	—
Mass (total)	—		

19 Bill of materials

24 positions P-01 … P-23B with material, process and mass (page 1 of 2)

ID	Description	Qty	Material	Manufacturing process	Mass [kg]
P-01	Rear flange with welded hub	1	EN AW-6061-T6	Laser cutting + turning + TIG welding	0.3285
P-02	Conical drum with 3 strap channels	1	EN AW-6061-T6	Flow forming / CNC turning + milling	0.2577
P-03	Hook-and-loop strapping bands	3	PP/PA webbing	Purchased, made up	0.0233
P-04	Front removable guard flange	1	EN AW-6061-T6	Laser cutting + CNC centring collar	0.2701
P-05	Axial retaining washer	1	C45E galvanised	Turning / stamping (DIN 988)	0.0106
P-06	Locking pin with pull ring	1	Galvanised steel	Standard part DIN 11023	0.0091
P-07	Main spindle with 3/8" drive stub	1	C45E +QT	CNC turning + milling + grinding	0.3954
P-08	Deep-groove ball bearing 6003-2RS	2	100Cr6 (SKF)	Standard part, DIN 625-1	0.0779
P-09	Stepped bearing housing	1	EN AW-6061-T6	CNC turning + drilling	0.1452
P-10	Bearing spacer sleeve	1	EN AW-6061-T6	Turned from tube	0.0163
P-11	Main chassis plate	1	EN AW-6061-T6	Laser cutting + deburring	0.2230
P-12	Reinforced guide arm (8 mm)	1	EN AW-6061-T6	CNC milling / waterjet	0.0547
P-13	Cantilever guide axle	1	C45E +QT	CNC turning + deep-hole drilling	0.1728
P-14	Deep-groove ball bearing 6002-2RS	2	100Cr6 (SKF)	Standard part, DIN 625-1	0.0600
P-15	Tape guide roller Ø38	1	POM-C	CNC turning	0.1053
P-16	4-lane tape guide carriage	1	POM-C	CNC milling / turning	0.0217
P-17	Leg strips, left & right	2	EN AW-6061-T6	Laser cutting + welding to P-18	0.0344
P-18	Cross member, hollow 20×20	1	EN AW-6061-T6	Sawing + drilling + counterboring	0.1134
P-19	Floor rails, left & right	2	EN AW-6061-T6	Sawing + drilling	0.2268
P-20	Foot plate (Z = 12…18)	1	EN AW-6061-T6	Milling / extruded profile	0.1258
P-21	Drill torque support	1	EN AW-6061-T6 + NBR	Bending + milling + NBR lining	0.1447
P-22	Valve strain-relief clamp	1	EN AW-6061-T6 + POM	CNC milling	0.0443
P-23	Elastomer feet & pads (A: 2 front, B: 4 rear)	6	NBR 70 Shore A	Purchased / moulded	0.0552
FAST	Standard fastener set (F-01…F-34)	34	Steel 8.8 galvanised / A2	Purchased standard parts	0.1810
Σ	**Total machine mass, dry**				**3.097**

REF-01…REF-03 are reference geometries for collision checks and are not purchased or manufactured. Mass excludes drill, battery and tape.

Bill of materials: dimensions and position

Main dimensions, bores/fits and installation envelope (page 2 of 2)

ID	Main dimensions [mm]	Bores & fits	Position [mm]	Mounted with
P-01	Ø220×3 disc + hub Ø44/Ø17×18	Hub bore Ø17.02 H7; cross hole Ø6.0 H11; 6× Ø5.5 on PCD R=38; 3 radial slots 16×58	X=[−18…3], Y=0, Z=150	P-02, P-07 · F-11…16, F-21
P-02	Ø104/Ø96.3 × 110, wall 2.0	6× Ø5.5 on PCD R=38; 3 axial channels 16×6 at 0°/120°/240°	X=[3…113], Y=0, Z=150	P-01, P-03, P-04 · F-11…16
P-03	15 × 380 × 1.5	Metal/PA buckle loop	X=[5…111] at 0°/120°/240°	P-02, REF-01
P-04	Ø220×3 + collar Ø30×1	Centring bore Ø17.20 H8; 6× Ø18; 3 slots 16×58	X=[113…117], Y=0, Z=150	P-02, P-05, P-07
P-05	Ø34 / Ø17.2 × 2.0	Bore Ø17.20 (+0.20)	X=[117…119], Y=0, Z=150	P-04, P-06, P-07
P-06	Shank Ø6.0 × 30, ring Ø17	Mates with Ø6.2 bore in P-07	X=122, Y=0, Z=[135…175]	P-05, P-07
P-07	Ø17.00 k5 × 212 + hex AF 9.5 × 36	Bearing seat Ø17.00 k5; shoulder Ø20; cross hole Ø6.0 (X=−9); pin hole Ø6.2 (X=122)	X=[−120…128], Y=0, Z=150	P-01, P-04…06, P-08, P-10 · F-21
P-08	d=17.00 × D=35.00 × B=10.00	Inner ring on Ø17 k5; outer ring in Ø35 H7	X=[−84…−74] & [−34…−24], Z=150	P-07, P-09, P-10
P-09	Tube Ø45×56, flange Ø68×4	2× bearing seat Ø35.00 H7 (10 deep); inner shoulder Ø31; 4× Ø5.5 on PCD R=28	X=[−84…−24], Y=0, Z=150	P-08, P-11 · F-01…04
P-10	Ø22.0 / Ø17.1 × 40.0	Bore Ø17.10 (+0.10); runout 0.02	X=[−74…−34], Y=0, Z=150	P-07, P-08A/B
P-11	150 × 170 × 4.0	Central Ø36; 4× Ø5.5 (PCD 28); 4× Ø5.5 (legs); 2× Ø5.5 (arm); 2 lightening windows 30×70	X=[−24…−20], Y=[−75…75], Z=[65…235]	P-09, P-12, P-17 · F-01…10
P-12	28 × 95 × 8.0	2× Ø5.5 (Z=212, 228); 1× Ø10.2 (Z=285)	X=[−32…−24], Y=[−14…14], Z=[205…300]	P-11, P-13 · F-09/10, F-22
P-13	M10×12 + Ø16×10 + Ø15.00 k5 × 142	Bearing seat Ø15.00 k5; hollow bore Ø8×120; thread M10×1.5	X=[−36…128], Y=0, Z=285	P-12, P-14 · F-22
P-14	d=15.00 × D=32.00 × B=9.00	Inner ring on Ø15 k5; outer ring in Ø32 H7	X=[−14…−5] & [109…118], Z=285	P-13, P-15
P-15	Ø38.0 / Ø26.0 × 132.0	2× bearing seat Ø32.00 H7 × 9.0 deep	X=[−14…118], Y=0, Z=285	P-14, P-16
P-16	5 guide collars Ø47/Ø38 × 2.0	Lane centres X = 19, 45, 71, 97 (lane width 26)	X=[4.5…110.5], Y=0, Z=285	P-15, REF-01
P-17	20 × 82 × 4.0	Each 2× Ø5.5 at Z=75 and Z=100	X=[−28…−24], Y=±[45…65], Z=[28…110]	P-11, P-18 · F-05…08
P-18	20 × 20 × 2.5 × 240	2× Ø6.5 (Y=±95), 2× Ø6.5 (Y=±30); flat counterbore on underside Z=8…12	X=[−34…−14], Y=[−120…120], Z=[8…28]	P-17, P-19, P-20, P-23B · F-17…20
P-19	20 × 20 × 2.5 × 240	Each 1× Ø6.5 (X=−24), 1× Ø5.5 (X=196), 2× Ø5.5 (strain relief)	X=[−34…206], Y=±[85…105], Z=[28…48]	P-18, P-22, P-23A · F-17/18, F-27…30
P-20	140 × 100 × 3.0 + 4 ribs 4×3	2× Ø6.5 (X=−24, Y=±30); 4× Ø5.5 (torque support); 2× Ø5.5 (pads)	X=[−154…−14], Y=[−50…50], Z=[12…18]	P-18, P-21, P-23B · F-19…26
P-21	60 × 60 × 93; handle fork 45 × 36	4× Ø5.5 base flange; 2× M5 clamp screws	X=[−235…−175], Y=[−30…30], Z=[15…108]	P-20, REF-02 · F-23…26
P-22	32 × 30 × 36; tape channel 26 × 4	2× Ø5.5 mounting to right floor rail; quick clamp	X=[42…74], Y=[−135…−105], Z=[28…64]	P-19R, REF-03 · F-27/28
P-23	A: Ø24/Ø6.0 × 28 · B: Ø24/Ø5.5 × 12	Through-bore Ø5.5, counterbored	A: X=196, Y=±95, Z=[0…28] · B: X=−24 (Y=±110) & X=−140 (Y=±40), Z=[0…12]	P-18, P-19, P-20 · F-29…34
FAST	24× M5, 5× M6, 1× M10×1.5 incl. washers & nuts	All bolt heads Z ≥ 12 mm above ground (REV-05)	See fastener register	All frame and rotor joints

20 Fastener register

34 fasteners F-01 … F-34 with position and mounting direction (page 1 of 2)

ID	Standard & thread	Size & head	XYZ centre [mm]	Axis & direction	Joins	Torque [Nm]
F-01	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=+19.8, Z=169.8 (45°)	+X (head at X=-28, nut X=-20)	P-09 + P-11	5.5
F-02	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=-19.8, Z=169.8 (135°)	+X (head at X=-28, nut X=-20)	P-09 + P-11	5.5
F-03	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=-19.8, Z=130.2 (225°)	+X (head at X=-28, nut X=-20)	P-09 + P-11	5.5
F-04	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=+19.8, Z=130.2 (315°)	+X (head at X=-28, nut X=-20)	P-09 + P-11	5.5
F-05	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=+55.0, Z=75.0	+X (head at X=-28, nut X=-20)	P-11 + P-17L	5.5
F-06	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=+55.0, Z=100.0	+X (head at X=-28, nut X=-20)	P-11 + P-17L	5.5
F-07	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=-55.0, Z=75.0	+X (head at X=-28, nut X=-20)	P-11 + P-17R	5.5
F-08	ISO 4017 M5x16	M5 x 16, hex head AF 8	X=-28..-16, Y=-55.0, Z=100.0	+X (head at X=-28, nut X=-20)	P-11 + P-17R	5.5
F-09	ISO 4017 M5x20	M5 x 20, hex head AF 8	X=-32..-16, Y=0.0, Z=212.0	+X (head at X=-32, nut X=-20)	P-11 + P-12	5.5
F-10	ISO 4017 M5x20	M5 x 20, hex head AF 8	X=-32..-16, Y=0.0, Z=228.0	+X (head at X=-32, nut X=-20)	P-11 + P-12	5.5
F-11	ISO 4017 M5x12	M5 x 12, hex head AF 8	X=0..10, Y=+32.9, Z=169.0 (30°)	+X (head at X=0, nut in cone X=6)	P-01 + P-02	5.5
F-12	ISO 4017 M5x12	M5 x 12, hex head AF 8	X=0..10, Y=0.0, Z=188.0 (90°)	+X (head at X=0, nut in cone X=6)	P-01 + P-02	5.5
F-13	ISO 4017 M5x12	M5 x 12, hex head AF 8	X=0..10, Y=-32.9, Z=169.0 (150°)	+X (head at X=0, nut in cone X=6)	P-01 + P-02	5.5
F-14	ISO 4017 M5x12	M5 x 12, hex head AF 8	X=0..10, Y=-32.9, Z=131.0 (210°)	+X (head at X=0, nut in cone X=6)	P-01 + P-02	5.5
F-15	ISO 4017 M5x12	M5 x 12, hex head AF 8	X=0..10, Y=0.0, Z=112.0 (270°)	+X (head at X=0, nut in cone X=6)	P-01 + P-02	5.5

Fastener register (continued)

Page 2 of 2 · all bolt heads stay at Z ≥ 12 mm above ground (REV-05)

ID	Standard & thread	Size & head	XYZ centre [mm]	Axis & direction	Joins	Torque [Nm]
F-16	ISO 4017 M5x12	M5 x 12, hex head AF 8	X=0..10, Y=+32.9, Z=131.0 (330°)	+X (head at X=0, nut in cone X=6)	P-01 + P-02	5.5
F-17	ISO 4762 M6x45	M6 x 45, socket head, hex key 5	X=-24.0, Y=+95.0, Z=12..53 (REV-05)	+Z (head recessed Z=8..12, nut Z=48)	P-18 + P-19L	9.5
F-18	ISO 4762 M6x45	M6 x 45, socket head, hex key 5	X=-24.0, Y=-95.0, Z=12..53 (REV-05)	+Z (head recessed Z=8..12, nut Z=48)	P-18 + P-19R	9.5
F-19	ISO 7380 M6x25	M6 x 25, button head, hex key 4	X=-24.0, Y=+30.0, Z=12..33 (REV-05)	+Z (head at Z=12, nut on top Z=28)	P-18 + P-20	9.5
F-20	ISO 7380 M6x25	M6 x 25, button head, hex key 4	X=-24.0, Y=-30.0, Z=12..33 (REV-05)	+Z (head at Z=12, nut on top Z=28)	P-18 + P-20	9.5
F-21	ISO 4762 M6x50	M6 x 50, fitted cross screw, hex key 5	X=-9.0, Y=-25..+25, Z=150.0 (REV-03)	+Y (across hub Ø44 and shaft Ø17)	P-01 + P-07	10
F-22	ISO 10511 M10x1.5	M10x1.5 lock nut AF 17	X=-36..-32, Y=0.0, Z=285.0 (REV-02)	-X (on threaded stub of P-13)	P-12 + P-13	35
F-23..26	4x ISO 7380 M5x16	4x M5 x 16, button head, hex key 3	X=-225..-185, Y=±24, Z=12..24 (REV-04)	+Z (up from foot plate Z=12)	P-20 + P-21	5.5
F-27..28	2x ISO 4762 M5x16	2x M5 x 16, socket head, hex key 4	X=48 & 68, Y=-105, Z=38.0	-Y (on right floor rail P-19R)	P-19R + P-22	5.5
F-29..34	6x ISO 4762 M5x16	6x M5 x 16, countersunk in rubber foot	X=196 (Y=±95), X=-24 (Y=±110), X=-140 (Y=±40)	+Z (deep in rubber foot Z>=8)	P-18 + P-19 + P-20 + P-23A	4

Washers and nuts per position: ISO 7089 washers and ISO 10511 lock nuts, as listed in the workbook. Torque values are guide values; blank torque means the workbook gives no value for that position.

21 Assembly instructions

Initial assembly sequence from the engineering report (steps 1–3)

Step 1 · Bolt the base frame (JNT-09)

Stand cross member P-18 with welded leg strips P-17L/R upright. Fit floor rails P-19L/R with 2× M6×45 (F-17/18) from below in +Z and foot plate P-20 with 2× M6×25 (F-19/20) from below in +Z. Then screw on the six elastomer feet P-23A/B (F-29…34), the torque support P-21 (F-23…26) and the strain-relief clamp P-22 (F-27/28).

Step 2 · Mount chassis plate and guide arm (JNT-07)

Fasten main chassis plate P-11 to leg strips P-17L/R with 4× M5×16 (F-05…08). Bolt reinforced guide arm P-12 to the upper end of the chassis with 2× M5×20 (F-09/10) *before* the rear flange goes on, so tool access in +X stays free.

Step 3 · Join the spindle bearing assembly (JNT-01…03)

Press rear bearing 6003-2RS (P-08A) and front bearing (P-08B) against the Ø31 mm inner shoulder in housing P-09. Insert spindle P-07 with spacer sleeve P-10 from the front (−X) until the Ø20 shoulder sits on the front bearing. Bolt housing P-09 to chassis P-11 with 4× M5×16 (F-01…04).

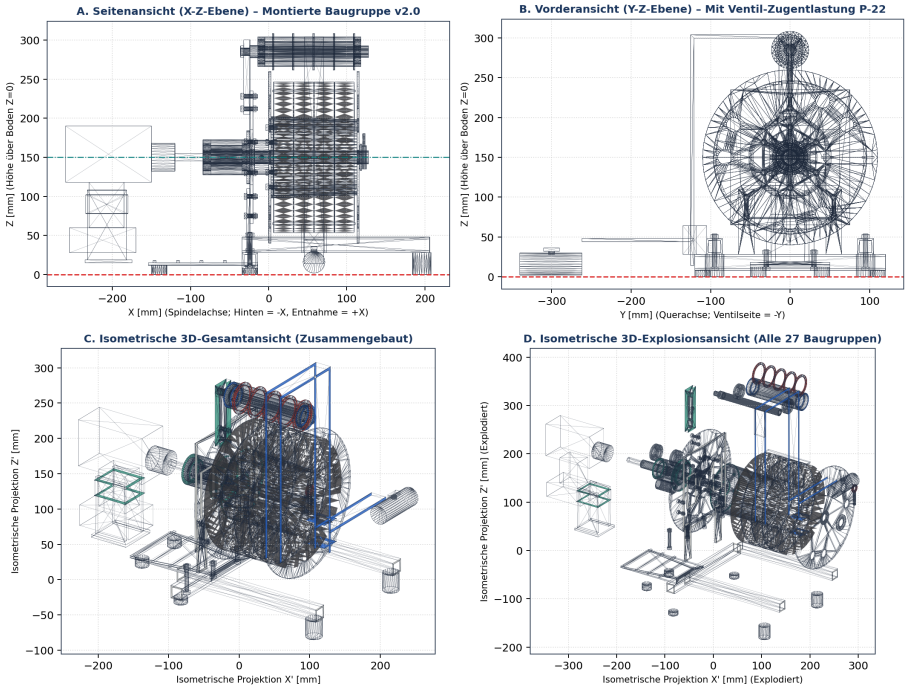


Figure 14 · Reference views of the assembled model for steps 1–3 (report figure).

Assembly instructions (continued)

Steps 4 and 5, plus exploded view

Step 4 · Mount drum and rear flange (JNT-04/05)

Bolt conical drum P-02 to rear flange P-01 with 6× M5×12 (F-11…16); the three strap channels must align exactly with the radial slots at 0°, 120° and 240°. Slide the assembly onto spindle P-07 and tighten the M6×50 cross bolt F-21 at X = −9.0 mm to 10 Nm.

Step 5 · Complete guide roller and quick-release (JNT-06/08)

Press two 6002-2RS bearings (P-14A/B) into roller P-15, fit 4-lane carriage P-16, slide the roller onto axle P-13 and secure the axle with M10×1.5 nut F-22 (35 Nm) on guide arm P-12. Insert the hook-and-loop straps P-03, push on front flange P-04 and washer P-05, and insert the locking pin P-06 at X = 122 mm.

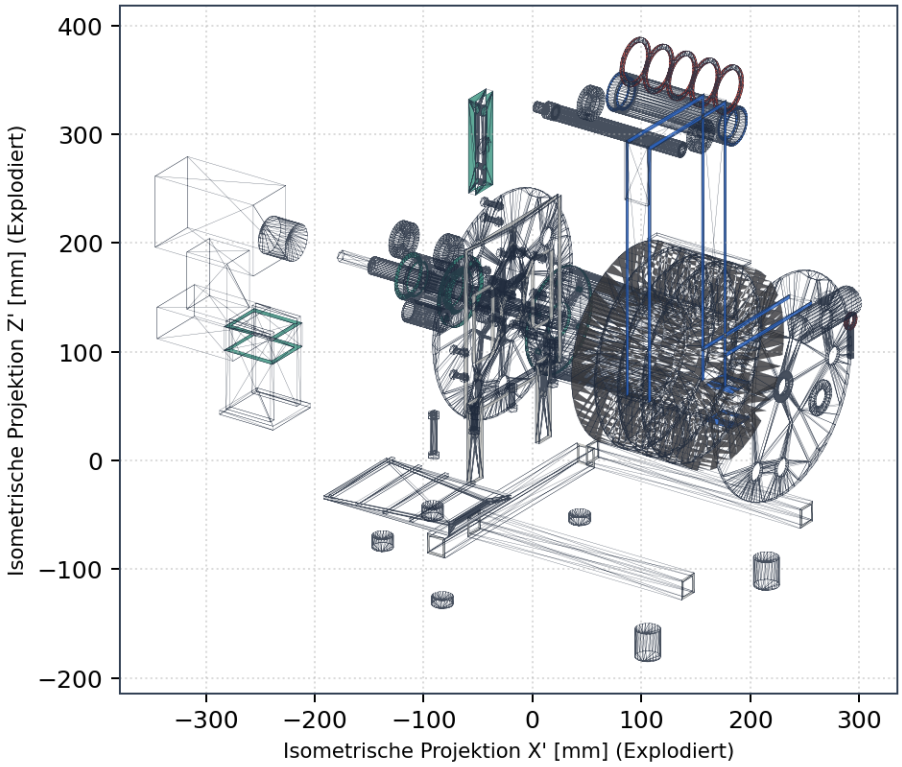


Figure 15 · Exploded isometric: drive side at left, drum and front flange at right. Drawing sheet 2 (page 11) adds item numbers and direction vectors.

Torque guide values (from the fastener register): M5 bolts about 5–5.5 Nm, M6 cross bolt 10 Nm, M10 axle nut 35 Nm. Check bearing seats with the gauges listed in the joint register on page 25.

22 Strapping and coil removal

Validated five-step sequence to take off a finished 40 m coil in +X without disconnecting the valve

Step 1 · Close the strain relief

After winding, lock the valve-side tape section in clamp P-22 so the valve carries 0 N.

Step 2 · Strap the coil at three points (REV-01)

Pass the three hook-and-loop straps P-03 (15 × 380 mm) through the 16 mm slots (0°, 120°, 240°) of rear flange P-01 and front flange P-04, and through the three 6 mm-deep drum channels of P-02 under the tape coil. Tighten from the outside.

Step 3 · Pull the locking pin

Pull pin P-06 (Ø6.0 at X = 122 mm) upward in +Z by its ring (Z = 172 mm).

Step 4 · Remove washer and front flange

Slide washer P-05 (X = 117…119) and the free front flange P-04 (X = 113…117, centring bore Ø17.20 H8, no bolts) along the spindle in +X.

Step 5 · Slide the strapped coil off in +X

Because of the 2° cone (radius 52.00 → 48.15 mm) the inner tape layer lifts 0.175 mm off the drum after only 5 mm of travel, so the coil slides off freely. The valve connection stays attached.

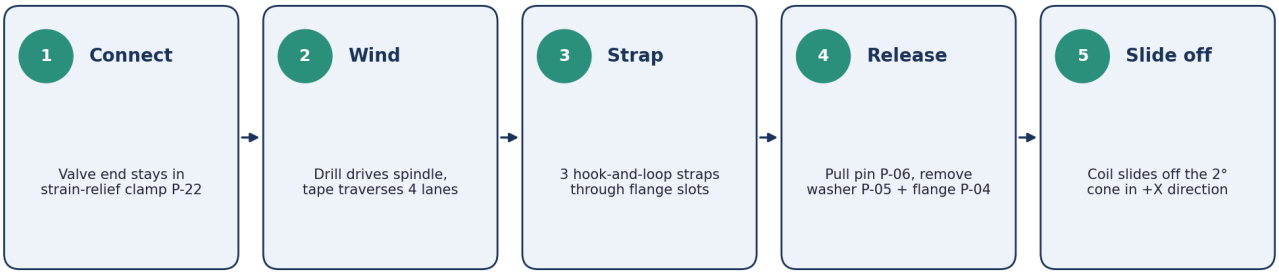


Figure 16 · Full operating sequence. The strapped bundle keeps its shape after removal.

Sheet 2 of the technical drawings (page 11) repeats this sequence beside the exploded view with direction vectors.

23 Operation and safety notes

How to use the winder within the verified limits

Before winding

- Fit the speed-reducing gearbox (12:1 to 16:1) between drill and spindle. Without it, the tape moves at up to 3.5 m/s (RISK-02).
- Set the drill to **1st gear** and the clutch to **stage 6–8** (about 8 Nm). Do not use 2nd gear.
- Seat the drill handle in the NBR-padded fork of support P-21 so the reaction torque goes into the frame.
- Start at the free field end of the tape in the drum core slot and wind toward the valve. Keep the valve end in clamp P-22 (RISK-01).

While winding

- Guide the tape by hand onto the roller; a steady pace of about 4 minutes per 40 m (21.7 rpm) matches the design.
- If the tape snags, the clutch must trip well before the 280 N tape breaking load (jam case 120 N).
- Keep hands out of the strap slots and between flange and frame while the drum turns.

After winding

- Strap first, then remove the pin, washer and front flange. Never pull the pin while the drum can still turn.
- Keep the tape coil clear of the spindle end when sliding it off along +X.

Limits that should not be exceeded

Quantity	Verified limit
Tape pull (jam, clutch stage 6)	120 N
Stall torque (design load for structure)	20 Nm
Tape length / coil radius	40 m / R_coil 96.58 mm (flange 110 mm)
Winding time window	3–5 min (17.4–28.9 rpm)
Dry machine mass	3.097 kg

24 Maintenance, materials and standards

Keeping the machine reliable and what it is made of

Maintenance

- The four ball bearings (2× SKF 6003-2RS, 2× SKF 6002-2RS) are sealed on both sides and lubricated for life.
- After each field use, brush dry soil out of the three drum channels (P-02) and the four lane collars (P-16).
- Lightly grease the cross hole of locking pin P-06.
- Check the M6×50 cross bolt F-21 (10 Nm) and the M10 axle nut F-22 (35 Nm) after the first hours of use.

Materials and standards used

Material / standard	Used for
EN AW-6061-T6 aluminium	Flanges, drum, chassis, housing, arm, frame profiles, clamps (density 2.70 g/cm³)
C45E +QT steel	Main spindle P-07 and guide axle P-13 (yield strength 430 MPa used in checks)
100Cr6 bearing steel · SKF 6003-2RS / 6002-2RS · DIN 625-1	Four sealed deep-groove ball bearings
POM-C	Guide roller P-15 and 4-lane carriage P-16
NBR 70 Shore A	Feet, pads and the torque-support lining
PP/PA webbing	Hook-and-loop straps P-03
Steel 8.8 galvanised / A2-70	ISO 4017 / 4762 / 7380 bolts, ISO 7089 washers, ISO 10511 lock nuts
ISO 286 fits · ISO 2768-mK	Shaft/housing fits (k5, H7, H8) and general tolerance on all drawing sheets

Reference tape and drive

Tape: ALLWIN GOLD flat drip tape, Allwin Pipes (Vijayamangalam, Erode, Tamil Nadu, India), 16 mm, 25 mm lay-flat, minimum dry bend radius 48.0 mm, breaking load about 280 N. Drive: INGCO CIDLI12206 12 V impact drill (0–400 / 0–1500 rpm, 20 Nm). Bearings: SKF 6003 (17×35×10 mm, C = 6.37 kN, C0 = 3.25 kN) and 6002 (15×32×9 mm, C = 5.85 kN, C0 = 2.85 kN).

25 Validation matrix

What was checked, how, and what is still open

A design is only as credible as the line it draws between “calculated”, “checked in CAD” and “tested on a bench”. This table keeps that line visible.

ID	Check	Method	Result in revision 2.0	Status
CHK-01	Unique part IDs, coordinates and BOM consistency	CAD and workbook as single source of truth	24 positions (P-01…P-23B), 34 fasteners, 11 joints identical in CAD, BOM, drawings and viewer	✓ Passed
CHK-02	Bearing, shaft and housing fits to DIN/ISO	SKF catalogue + calculation	6003-2RS (17×35×10) on Ø17.00 k5 / Ø35.00 H7; 6002-2RS (15×32×9) on Ø15.00 k5 / Ø32.00 H7	✓ Passed
CHK-03	Collision-free, ground clearance, coil removal in +X	3D CAD geometry check	Min. 12.0 mm clearance (REV-05); 3 strap slots and 2.0° cone allow free pull in +X	✓ Passed
CHK-04	Mass ≤ 5.00 kg without drill and battery	Volume-based mass balance	3.097 kg dry (1.903 kg reserve)	✓ Passed
RISK-01	Stationary valve vs. rotating drum	Kinematic analysis + field test	Clamp P-22 isolates 100% of tape force; winding must run from the free field end toward the valve (or fit a rotary union)	⚠ Field test open
RISK-02	Speed reduction for 3–5 min winding (21.7 rpm)	Drive calculation + test bench	INGCO drill runs 0–400 rpm, direct drive about 18× too fast; planetary gear 12:1…16:1 required before series release	⚠ Release gate open

Not claimed: no series or manufacturing readiness is claimed while the two physical tests are pending. For prototype build and bench testing the design is fully dimensioned, collision-checked and analytically validated.

26 Open risks and mitigation

The two release gates in detail

RISK-01 · Stationary valve end versus rotating drum

Problem: a tape end fixed to both the spinning core and the stationary valve is twisted once per revolution (86.82 times). Mitigation in v2.0: wind from the free end toward the valve and park the final 0.8 m loop in strain-relief clamp P-22; the clamp has a 26 × 4 mm channel, takes 120 N and leaves 0 N at the valve (acceptance: pull test 120 N → 0 N). Remaining action: run the field procedure on real soil, or retrofit a rotary union if the valve must stay connected throughout.

RISK-02 · Drive speed

Problem: the drill delivers 0–400 rpm in 1st gear; the spindle should turn at 17.4–28.9 rpm. Mitigation: a compact planetary gearbox between chuck and spindle (12:1 for 3 min, 16:1 for 4 min, 20:1 for 5 min at about 347 rpm drill speed), clutch stage 6–8, 2nd gear locked out. Remaining action: select the gearbox and verify temperature and torque on a test bench.

Assumptions still to confirm

Parameter	Assumed value	How to confirm
Effective tape thickness t_eff	1.5 mm (single wall 0.7 mm + emitter 2.2–3.0 mm averaged)	Measure the real tape with emitters
Packing factor η	0.70 (conservative)	Wind a test coil and measure R_coil; worst case still clears the flange by 5.2 mm
Tape pull in winding	35 N nominal, 120 N max	Pull-scale test on dry field soil
Clutch-limited torque	about 8 Nm at stage 6	Bench test of the drill with gearbox
Tape lay-flat width	25.0 mm (24.5–25.5)	Caliper check on delivered tape

27 Proposed prototype test plan

Next steps that turn the validated design into a validated machine (proposal, not yet done)

#	Test	Pass criterion (from this design)	Related
1	Measure bearing seats on spindle, housing and axle	Ø17.002…17.010 on spindle; Ø35 H7 and Ø32 H7 by go/no-go gauge; Ø15.00 k5 on axle	JNT-01/02/08
2	Flange runout and slot alignment	Runout < 0.15 mm; slots match drum channels	JNT-05
3	Quick-release time	Pin, washer and flange removed in < 5 s without tools	JNT-06
4	Stall-torque test at the cross bolt	20 Nm without slip	JNT-04
5	Torque-support test with the drill	20 Nm reaction held; battery change still possible	JNT-10
6	Strain-relief pull test	120 N on the tape gives 0 N at the valve	JNT-11
7	Ground-clearance gauge	≥ 12.0 mm under all fasteners	JNT-09
8	Wind 40 m of real tape with gearbox	3–5 min; $R_{coil} \leq 104.8$ mm; no tape damage at $R \geq 48$ mm	RISK-02, CAP-04
9	Field run of valve procedure	No twist or load at the valve end	RISK-01

Results of these tests will go into the next revision (v2.1). Until then the document only claims what the model and the calculations prove.

28 Manufacturing summary

Processes and purchased parts grouped by how they are made

Group	Positions	Typical processes
Laser-cut aluminium plate	P-01, P-04, P-11, P-17	Laser cutting, deburring, welding (hub to rear flange P-01; legs to cross member), CNC centring collar
Turned / flow-formed aluminium	P-02, P-09, P-10	Flow forming or CNC turning with milling; turning from tube; drilling
Milled aluminium	P-12, P-20, P-22	CNC milling or waterjet; milled or extruded plate; NBR/POM inserts
Hollow profile 20×20×2.5	P-18, P-19	Sawing, drilling, counterboring
Ground steel shafts	P-07, P-13	CNC turning, milling or deep-hole drilling, then grinding to k5
POM-C turned parts	P-15, P-16	CNC turning and milling
Bent and lined	P-21	Bending, milling, NBR lining
Purchased standard parts	P-03, P-05, P-06, P-08, P-14, P-23, FAST	Webbing straps, DIN 988 washer, DIN 11023 pin, SKF bearings, NBR feet, ISO fasteners

Tolerances and surface

- General tolerance ISO 2768-mK on all drawing sheets.
- Ground bearing seats: Ø17.00 k5 (+0.002/+0.010 mm) on the spindle, Ø15.00 k5 on the guide axle, Ra 0.8 µm.
- Housing seats Ø35.00 H7 (+0/+0.025 mm); roller seats Ø32.00 H7; front flange centring bore Ø17.20 H8.
- Hub bore Ø17.02 H7 (+0.018/0) with cross bore Ø6.0 H11 finished after welding.
- Cone: $R(x) = 52.00 - (3.85/110)\cdot(x - 3.0)$ on $X = 3.0 \ldots 113.0$ mm, wall 2.0 mm.

Manufacturing drawings for the rotating group, spindle and bearings, guide, and frame are on sheets 3–6 (pages 12–15).

29 Digital deliverables and CAD source

What is in the project package

File / folder	Content
NEox_Compact_40M_Interactive_CAD.html	Interactive 3D CAD and 2D kinematic simulator (60 FPS, coaxial rotation, STL/CSV/JSON export)
NEox_Compact_40M_Project.html	Project portal with kinematics simulator, live calculation model, BOM and drawings
NEox_Compact_40M_Engineering_Report_DE.pdf	12-page engineering and design report (German)
NEox_Compact_40M_Calculations_BOM_DE.xlsx	12-sheet calculation workbook with evaluated formulas, BOM, fastener and joint registers
Drawings/...Technical_Drawings_DE.pdf	6 drawing sheets: assembly, exploded, rotating group, spindle, guide, frame
CAD/STEP	Assembly + 27 part files (ISO 10303-21 AP214)
CAD/STL	Complete assembly + 27 binary STL parts
CAD/GLB	Assembled and exploded 3D scenes
CAD/Parametric_Scripts	CadQuery, OpenSCAD and Python mesh-kernel scripts that rebuild the model
Documentation	Change log and test report (German) and the YAML master specification

Excerpt of the parametric CadQuery script

```
# Master-Parameter v2.0
ZC = 150.0
R_FLANGE = 110.0
T_FLANGE = 3.0
R_CORE_BACK = 52.0
R_CORE_FRONT = 48.15
L_CONE = 110.0
D_SPINDLE_K5 = 17.00
D_BEARING_6003_H7 = 35.00
D_GUIDE_SHAFT_K5 = 15.00
D_BEARING_6002_H7 = 32.00

def build_rear_flange_p01():
    # Scheibe Ø220x3 bei X=0..3 + Nabe Ø44x18 bei X=-18..0 + 3x Umreifungsschlitz 16x58 (REV-01)
    disc = cq.Workplane("YZ").workplane(offset=0.0).circle(R_FLANGE).extrude(T_FLANGE)
    hub = cq.Workplane("YZ").workplane(offset=-18.0).circle(22.0).extrude(18.0)
    part = disc.union(hub).faces("<X").workplane().circle(D_SPINDLE_K5/2.0 + 0.01).cutThruAll()
    for deg in [0, 120, 240]:
        part = (part.faces(">X").workplane()
                .transformed(rotate=(0, 0, deg))
                .center(74.0, 0).slot2D(58.0, 16.0, 0).cutThruAll())
    return part.translate((0, 0, ZC))

def build_conical_drum_p02():
    # Konischer Trommelmantel X=3..113, R=52,00 -> 48,15 mm, Wandstärke 2,0 mm
    outer = cq.Solid.makeCone(R_CORE_BACK, R_CORE_FRONT, L_CONE, cq.Vector(3, 0, ZC), cq.Vector(1, 0, 0))
    inner = cq.Solid.makeCone(R_CORE_BACK - 2.0, R_CORE_FRONT - 2.0, L_CONE, cq.Vector(3, 0, ZC), cq.Vector(1, 0, 0))
    return cq.Workplane(obj=outer.cut(inner))
```

Master parameters (zero position at X = 0, spindle at Z = 150) are the same numbers used in the drawings and calculations, so changing one value regenerates a consistent model.

30 Highlights and contact

A short summary for reviewers

Metric	Value
Design iterations	v1.0 baseline → v2.0 with 7 logged changes (REV-01…07)
Conflicts found and solved	5 (bearing fits ×2, cone geometry, one-tape model, ground clearance)
Parts / fasteners / joints	24 BOM positions · 34 fasteners · 11 joints
Model files	28 STEP, 28 STL, 2 GLB, 3 parametric scripts
Drawing sheets	6 manufacturing and assembly sheets
Calculation workbook	12 sheets with evaluated formulas
Mass	3.097 kg dry vs. 5.0 kg target
Smallest verified safety factor	5.80 (spindle at cross bore, 20 Nm stall)
Open items	2 gates: valve kinematics field test and drive gearbox

What I would stress in an interview

- I found the contradictions in my own first version (a 17 mm bearing on a 16.8 mm shaft, a 6002 bearing that cannot fit a Ø16 axle) and fixed them with numbers, not guesses.
- I separate what is calculated, what is checked in CAD and what still needs a physical test, and I say so in the document.
- One master set of parameters drives the CAD, the drawings, the workbook and the viewers, so the documentation cannot drift apart.

Navaneeth

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neoxey.github.io/portfolio
Open to internships
CAD files, workbook and drawing sheets on request